When Can Safe Controllers Adapt?
Information before Commitment

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Abstract

Safe adaptive control is online adaptation under a safety guarantee on the learning trajectory itself. The controller may use any causal, history-dependent rule and act differently across environments as data arrive. Only its safety guarantee is uniform: the same rule must satisfy it under every initially plausible model. Performance is measured against a safe oracle that knows the realized model.

Many finite-time analyses assume persistent excitation of the uniformly safe closed loop, so the data distinguish every pair of models requiring different control decisions. Under that assumption, feasibility is already settled; only the rate remains. We ask instead: *Do the safety constraints permit such an informative experiment at all?*

While an alternative remains plausible, the controller must preserve a safe continuation under it. We call the first action that forecloses such a continuation *commitment*. Chance safety allows commitment only on an event rare under the alternative, and the evidence must arrive beforehand: the observation generated by the committing action is too late. We define *precommitment information* as the KL divergence between learner-visible laws stopped before commitment.

Our main result is a causal reduction. The commitment rule determines three system-dependent quantities: (a) the probability that safety permits commitment under the alternative, (b) the target-side cost of remaining noncommittal, (3) and the information available when the decision is made. Bounded precommitment information therefore leaves a fixed fraction of the oracle gap unavoidable. If the gap is $\Omega(T)$, every uniformly safe policy has linear regret. We establish the obstruction in a constrained linear system with quadratic regulation cost. With multiple alternatives, finite information limits how small a certified candidate list can be. We also prove recovery when safe experiments are repeatable, persistently informative, and return to a common continuation state, and derive semidefinite upper certificates for deterministic linear-Gaussian systems.

1 Introduction

The safe adaptive control problem is regret minimization over uniformly safe policies. A causal policy must satisfy the safety specification under every model not yet ruled out, and its performance is measured against the safe oracle that knows the true model. The difficulty is exploration. The policy can learn only through actions that remain safe under every plausible model, yet it must reach the oracle's model-specific behavior from inside that set.

Many finite-time analyses impose conditions under which this common safe regime remains informative. Excitation can be injected, or process noise and safety-induced state variation keep the closed loop informative [1, 2, 3, 4, 5]. Safety then limits where the system may go, but not the decision-relevant information it can acquire. Under such conditions, familiar adaptive-control rates can be recovered.

We ask what can be said when the constraints also limit the experiment. This is not merely a slower estimation problem. An action that distinguishes models may consume safety margin under a model that is still plausible. The admissible experiment set is then endogenous: it depends on the margin already spent and on the models not yet falsified. The model-specific action sought by adaptation is unavailable until

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the evidence supporting it has been collected. This creates an ordering constraint: evidence must precede commitment, while the same margin that commitment consumes may also limit the available evidence. The full controlled trajectory is therefore the wrong statistical experiment: it can contain evidence generated by the very action that already required the model-specific decision. The question is no longer only how fast uncertainty shrinks, but whether enough evidence can be collected before the first action that forecloses a safe continuation under an alternative. We call the evidence available before that action *precommitment information.* Bounded precommitment information certifies an unavoidable oracle gap. Persistent information generated by repeatable safety-preserving experiments supports recovery.

Guarantee or exclude. With uncertainty unresolved, a uniformly safe controller has two options. It can guarantee: remain in the regime that is feasible under every plausible model and accept the value this assures. In our lower-bound examples, this regime already attains the max–min value. Or it can exclude: once the evidence permits, take an action as if an alternative were no longer in play. The exclusion is statistical, not literal. The safety guarantee remains uniform over the initial model class; chance safety permits an action that is unsafe under an alternative only on an event whose probability under that alternative is at most η_T. Making commitment likely under the target while keeping it this rare under the alternative requires order $\log(1/\eta_T)$ nats of evidence, and that evidence must be available before the committing action is chosen. In the lower-bound examples, adaptation beyond the max–min value is precisely such an exclusion. The max–min criterion does not distinguish the two options once the worst-case value is fixed. Oracle regret does.

Oracle regret. Let the unknown environment be $\theta \in \Theta_0$. Over a horizon T, let $\Pi^T_{\text{safe}}(\Theta_0)$ be the causal policies whose probability of any safety violation is at most η_T under every $\theta \in \Theta_0$. Let $J^\star_T(\theta)$ be the value of the safe oracle that knows θ, and define

$$R_T(\pi;\theta) := J^\star_T(\theta) - J_T(\pi;\theta).$$

We study

$$\mathfrak{R}_T(\Theta_0) := \inf_{\pi \in \Pi^T_{\text{safe}}(\Theta_0)} \sup_{\theta \in \Theta_0} \big[J^\star_T(\theta) - J_T(\pi;\theta)\big], \tag{1}$$

and ask when it is $o(T)$. More generally, if the oracle advantage itself is $G_T = o(T)$, we ask whether the learner can close all but $o(G_T)$ of that gap.

Commitment. Fix a target environment θ and an incompatible alternative θ'. A commitment rule specifies, at each time t, a predictable set $\mathcal{D}_t^{\theta|\theta'}$ of history–action pairs. The induced commitment time and event are

$$\tau^\pi := \inf\big\{t \le T : (H_{t-1}, U_t) \in \mathcal{D}_t^{\theta|\theta'}\big\}, \qquad A^\pi := \{\tau^\pi \le T\}.$$

In deterministic constrained systems, commitment is the first action after which no hard-safe continuation remains under θ'. Noncommitment is therefore preservation of recursive feasibility under the alternative. Because the commitment decision is made from (H_{t-1}, U_t), the observation produced by U_{τ^π} arrives too late to justify that action.

The commitment rule is the fundamental object. It simultaneously determines how often safety permits commitment under the alternative, what noncommitment costs under the target, and how much information is available when the decision is made. Safety determines how often commitment is allowed under the alternative. Value determines the regret cost of not committing under the target. Information is measured on the learner-visible record stopped before commitment. The controller need not identify the full model; it needs only to resolve alternatives that require different safe decisions. This connects to sequential design and controlled sensing [6, 7, 8], with the additional restriction that safety controls which experiments are admissible and whether their observations arrive before commitment.

Precommitment lower bound. Let

$$\alpha_T(\theta,\theta') := \sup_{\mu \in \Pi^T_{\text{safe}}(\{\theta,\theta'\})} \mathbb{P}^\mu_{\theta'}(A^\mu)$$

be the largest commitment probability allowed under the alternative. If commitment forces a violation under θ', then $\alpha_T(\theta,\theta') \le \eta_T$. For $q \in [0,1]$, define the noncommitment regret profile

$$\mathcal{G}_T(\theta,\theta';q) := \inf_{\substack{\mu \in \Pi^T_{\mathrm{safe}}(\{\theta,\theta'\}) \\ \mathbb{P}^\mu_\theta((A^\mu)^c) \ge q}} R_T(\mu;\theta).$$

It is the least target regret among uniformly safe policies that do not commit with probability at least q.

Let $I_{\mathrm{pre}}(T;\theta,\theta')$ be the largest KL divergence between the stopped learner-visible laws under θ and θ'. Since A^π is determined by the stopped record, data processing links precommitment information to the two commitment probabilities. Theorem 1 gives an exact binary-KL bound. A simpler consequence is

$$R_T(\pi;\theta) \ge \mathcal{G}_T\left(\theta,\theta';\left[\frac{1}{2}e^{-I_{\mathrm{pre}}(T;\theta,\theta')} - \alpha_T(\theta,\theta')\right]_+\right). \tag{2}$$

If a system-specific calculation gives

$$\mathcal{G}_T(\theta,\theta';q) \ge [G_T q - \zeta_T]_+,$$

then

$$R_T(\pi;\theta) \ge \left[G_T\left(\frac{1}{2}e^{-I_{\mathrm{pre}}(T;\theta,\theta')} - \alpha_T(\theta,\theta')\right) - \zeta_T\right]_+. \tag{3}$$

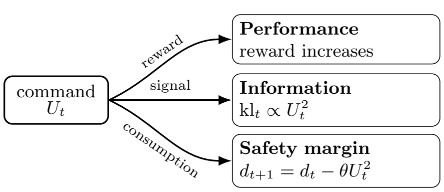
Thus bounded precommitment information leaves a nonvanishing fraction of the oracle gap. If $G_T = \Omega(T)$, every uniformly safe policy has linear oracle regret. At the standing risk level $\eta_T = T^{-2}$, the explicit bound remains positive until the stopped KL reaches $2\log T - \log 2$. Driving the target noncommitment probability to zero therefore requires evidence of order $2\log T$, which a bounded profile cannot supply.

Finite-headroom example. A simple example (see Figure 1) shows why waiting need not help. Over T rounds, a controller chooses commands[1] $U_t \in [0,U]$. The command earns reward but consumes thermal headroom:

$$d_{t+1} = d_t - \theta u_t^2, \qquad d_t \ge 0.$$

After applying u_t, the controller observes

$$y_t = \theta u_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0,\sigma^2).$$



Controller	Reward
Healthy oracle	UT
Worn-safe max–min baseline	$\min\{UT, \sqrt{Td_0/\theta_{\mathrm{high}}}\}$
Uniformly safe learner	may exceed the baseline only after evidence

Evidence required for error of order $\eta_T = T^{-2}$: order $\log(1/\eta_T) = 2\log T$
Evidence available before leaving the worn budget: $\theta_{\mathrm{high}} d_0/(2\sigma^2)$ — independent of the horizon

Figure 1: **Finite headroom.** A command affects reward, information, and thermal headroom. The evidence needed to leave the worn-safe regime grows with the horizon, while the evidence available before leaving it is bounded.

[1]The restriction to nonnegative commands is expositional, not structural. One may instead take $U_t \in [-U,U]$, reward $r_t = |U_t|$, and the same thermal dynamics and sensor; both safety expenditure and Gaussian information depend on U_t^2, so the sign is immaterial. A signed aeroelastic realization replaces these by a noisy flutter signal $Y_t = \theta U_t^2 + \xi_t$ and a fatigue budget $d_{t+1} = d_t - \theta^2 U_t^4$. Then both accumulated fatigue and pairwise information depend on $\sum_t U_t^4$, and the same matched-order argument applies.

Take $r_t = u_t$ and compare a healthy device $\theta_{\text{low}} = 0$ with a worn device $\theta_{\text{high}} > 0$. The healthy oracle uses $u_t = U$ in every round. Let τ^π be the first time the learner leaves the worn-device budget. Before commitment, safety under the worn model gives

$$\sum_{t<\tau^\pi} u_t^2 \leq \frac{d_0}{\theta_{\text{high}}}.$$

The stopped Gaussian KL therefore satisfies

$$I_{\text{pre}}(T; \theta_{\text{low}}, \theta_{\text{high}}) \leq \frac{(\theta_{\text{high}} - \theta_{\text{low}})^2}{2\sigma^2} \frac{d_0}{\theta_{\text{high}}} = \frac{\theta_{\text{high}} d_0}{2\sigma^2}, \tag{4}$$

which is independent of T. The required evidence grows as $2\log T$, but the available evidence does not grow at all. The healthy oracle gap is $\Theta(T)$, so the lower bound gives linear regret.

Recovery. An unbounded information profile alone does not prove recovery. One safe policy must generate the information and then enter the selected operating regime without losing feasibility. We give a sufficient condition based on a repeatable safety-preserving experiment. In the braking example, maximal braking is both safe and informative, and a confidence-based controller achieves $O(\sqrt{T \log T})$ regret.

Contributions.

1. **Commitment and precommitment information.** We define a predictable commitment rule for controlled systems. The same rule determines the safety allowance under the alternative, the noncommitment regret profile under the target, and the stopped information.

2. **Causal reduction.** We reduce safe oracle attainability to three quantities induced by the same commitment rule: the alternative-side commitment allowance, the target-side noncommitment regret profile, and the information in the learner-visible record available at the decision. This gives the exact binary-KL bound in Theorem 1. We extend the result to continuous classes and use a list-Fano bound to quantify partial specialization among multiple alternatives.

3. **Examples and recovery.** We prove linear regret in a constrained linear system with quadratic regulation cost and an unknown margin-consuming input direction. An unknown-disturbance-location problem yields a certified safe candidate list. A separate alignment condition gives oracle recovery at rate $O(\sqrt{T \log T})$ when safe experiments are repeatable and informative.

4. **Computation.** For deterministic linear-Gaussian systems, we formulate finite-horizon precommitment information as a QCQP and derive SDP upper certificates. Numerical studies show how replenishment, sensing, and unknown-location structure change the attainable level of adaptation.

2 Setup

We define the controlled interaction, uniform safety, the parameter-aware oracle, and commitment.

2.1 General controlled interaction

Fix a horizon T and an uncertainty set $\Theta_0 \subseteq \Theta$. A state-space representation, when useful, is

$$X_{t+1} = f_{\theta,t}(X_t, U_t, W_{t+1}), \qquad Z_t = h_{\theta,t}(X_t, U_t, V_t), \qquad t = 1, \ldots, T, \tag{5}$$

where X_t is the physical state, U_t is the chosen control or sensing action, Z_t is the learner-visible datum, and (W_t, V_t) are noises. Uppercase denotes random variables. The learner sees

$$H_{t-1} := (Z_1, U_1, \ldots, Z_{t-1}, U_{t-1})$$

before choosing $U_t \sim \pi_t(\cdot \mid H_{t-1})$. Equivalently, under θ, the next learner-visible datum has conditional law

$$Z_t \mid (H_{t-1}, U_t) \sim Q_{\theta,t}(\cdot \mid H_{t-1}, U_t). \tag{6}$$

The kernels $Q_{\theta,t}$ are induced by the dynamics and observation model. The lower bound uses only these controlled observation laws; it does not require linearity, Gaussian noise, or Markov representation in (5).

Let $\mathsf{Safe}_{\theta,T}$ be the event that all state, input, resource, and operational constraints hold through horizon T under θ. For state constraints, it includes the post-action state X_{T+1} generated by U_T. Let $\overline{P}^{\pi}_{\theta,T}$ denote the full physical trajectory law, including latent safety variables, and let $P^{\pi}_{\theta,T}$ denote the learner-visible marginal. Safety probabilities are taken under $\overline{P}^{\pi}_{\theta,T}$; KL divergences are taken on learner-visible records.

Definition 1 (Uniform chance safety)**.** For a risk level $\eta_T \in [0, 1)$, a policy is *chance-η_T safe* under θ if

$$\overline{P}^{\pi}_{\theta,T}(\mathsf{Safe}^c_{\theta,T}) \le \eta_T.$$

It is *uniformly chance-safe* on Θ_0 if the same causal policy satisfies this inequality for every initially plausible environment:

$$\Pi^T_{\text{safe}}(\Theta_0) := \bigcap_{\theta\in\Theta_0} \left\{\pi : \overline{P}^{\pi}_{\theta,T}(\mathsf{Safe}^c_{\theta,T}) \le \eta_T\right\}. \tag{7}$$

Hard safety is the special case $\eta_T = 0$. In the lower-bound examples, $\eta_T \to 0$, often $\eta_T = T^{-2}$. When reward is bounded, the contribution of violation paths to expected value then vanishes.

Remark 1 (Why safety is uniform). The policy must be certified before the true environment is known. Uniformity therefore applies to the safety guarantee, not to the realized behavior: a causal policy may use observations and act differently under different models. Every data-dependent decision inside the policy, including a decision to commit, remains covered by the guarantee in (7).

Remark 2 (Uniform safety and max–min control). The class $\Pi^T_{\text{safe}}(\Theta_0)$ contains general adaptive policies; a max–min policy may also be adaptive. The distinction in this paper is the performance criterion: safety is required uniformly over Θ_0, whereas performance is measured by regret to the oracle for the environment that is actually realized.

Let $\mathcal{R}_T$ denote realized reward through horizon T. We assume that its expectation and the oracle value below are finite. The main theorem imposes no pathwise reward ceiling. A finite gross bound is used only in Lemma 2 to evaluate the noncommitment regret profile in deterministic examples. The value of a policy π in environment θ is

$$J_T(\pi; \theta) := \mathbb{E}^{\pi}_{\theta}[\mathcal{R}_T]. \tag{8}$$

The parameter-aware safe oracle knows θ and is constrained only by safety under that environment:

$$J^{\star}_T(\theta) := \sup_{\pi\in\Pi^T_{\text{safe}}(\{\theta\})} J_T(\pi; \theta). \tag{9}$$

Regret is

$$R_T(\pi; \theta) := J^{\star}_T(\theta) - J_T(\pi; \theta). \tag{10}$$

Because $\Pi^T_{\text{safe}}(\Theta_0) \subseteq \Pi^T_{\text{safe}}(\{\theta\})$ for every $\theta \in \Theta_0$, the regret of a uniformly safe policy is nonnegative.

Definition 2 (Safe oracle attainability)**.** Let

$$\mathfrak{R}_T(\Theta_0) := \inf_{\pi\in\Pi^T_{\text{safe}}(\Theta_0)} \sup_{\theta\in\Theta_0} R_T(\pi; \theta).$$

For a positive comparison scale a_T, the oracle is safely attainable at scale a_T if $\mathfrak{R}_T(\Theta_0) = o(a_T)$. Average-regret learnability corresponds to $a_T = T$.

The scale matters: a lower bound of order $\sqrt{T}$ can rule out closure of a $\Theta(\sqrt{T})$ oracle advantage without ruling out $o(T)$ average regret.

2.2 Commitment and information before commitment

Fix an ordered pair $(\theta, \theta') \in \Theta_0^2$. The first environment is the target: leaving the common regime may be valuable there. The second is an incompatible alternative: the same decision is not safely recoverable there. These roles are relative to the decision under study; they do not impose a global ordering on the model class.

In the deterministic constrained systems used below, let $\mathcal{U}_t^{\mathrm{viab}}(h; \vartheta)$ denote the actions available after history h for which at least one continuation remains hard-safe under ϑ through the horizon.

Definition 3 (Pairwise commitment)**.** For the ordered pair (θ, θ'), define

$$\mathcal{D}_t^{\theta|\theta'} := \{(h, u) : u \notin \mathcal{U}_t^{\mathrm{viab}}(h; \theta')\}. \tag{11}$$

For a policy π, let

$$\tau^\pi := \inf\{t \le T : (H_{t-1}, U_t) \in \mathcal{D}_t^{\theta|\theta'}\}, \tag{12}$$

with $\tau^\pi = T + 1$ if the set is empty, and define

$$A^\pi := \{\tau^\pi \le T\}. \tag{13}$$

Thus commitment is the first action after which no hard-safe continuation remains under the incompatible alternative. Because membership in $\mathcal{D}_t^{\theta|\theta'}$ is determined by (H_{t-1}, U_t), the commitment decision is made before the observation Z_t generated by that action is available. For a general stochastic model, the same stopping-time construction may be used with an application-specific predictable set $\mathcal{D}_t^{\theta|\theta'}$; its safety consequence is then quantified by bounding the commitment allowance of Definition 5 below. No general chance-constrained viability operator is needed for the lower bound.

Finite-headroom commitment. Recall from Section 1 the controller chooses $u_t \in [0, U]$, receives reward $r_t = u_t$, and the safety margin evolves as $d_{t+1} = d_t - \theta u_t^2$, $d_1 = d_0 > 0$. For an ordered pair $0 \le \theta_b < \theta_d$, write $B_d := B_{\theta_d} = d_0/\theta_d$. Let

$$E_{t-1}(h) := \sum_{s=1}^{t-1} u_s^2.$$

Since future commands can be set to zero, for $\vartheta > 0$,

$$\mathcal{U}_t^{\mathrm{viab}}(h; \vartheta) = \left[0, \min\left\{U, \sqrt{\left(\frac{d_0}{\vartheta} - E_{t-1}(h)\right)_+}\right\}\right]. \tag{14}$$

For the ordered pair (θ_b, θ_d), commitment is therefore the first exit from the dangerous-device budget:

$$\mathcal{D}_t^{\theta_b|\theta_d} = \{(h, u) : E_{t-1}(h) + u^2 > B_d\}. \tag{15}$$

3 Fundamental Pre-Commitment Limits

The lower bound uses three quantities: commitment probability under the alternative, the regret cost of noncommitment under the target, and the information available before commitment. We define each quantity directly.

3.1 Binary target versus incompatible alternative

The committing action is chosen before its observation is seen, so the record on which information is measured should contain the committing action but not its outcome. We therefore use a stopping time. Let $\mathcal{F}_t := \sigma(H_{t-1}, U_t)$ be the σ-algebra generated by the learner-visible history and the time-t action but *not* the time-t

observation Z_t. Because whether $(H_{t-1}, U_t) \in \mathcal{D}_t^{\theta|\theta'}$ is decided by U_t before Z_t is seen, the commitment set is $\mathcal{F}_t$-measurable, and hence τ^π is an $\{\mathcal{F}_t\}$-stopping time.

Let $\mathcal{F}_{\tau^\pi}$ be the associated stopped σ-algebra. The stopped record

$$S_T^\pi := \Big(\tau^\pi \wedge (T+1),\ H_{(\tau^\pi-1)\wedge T},\ \widetilde{U}^\pi\Big), \qquad \widetilde{U}^\pi := \begin{cases} U_{\tau^\pi}, & \tau^\pi \le T, \\ \dagger, & \tau^\pi = T+1, \end{cases} \tag{16}$$

generates $\mathcal{F}_{\tau^\pi}$; the placeholder $\dagger$ marks the absence of a committing action. Let $P_{\theta,T}^{\pi,\mathrm{pre}}$ denote the law of S_T^π under θ, i.e. the restriction of the physical law to $\mathcal{F}_{\tau^\pi}$.

The commitment event $A^\pi = \{\tau^\pi \le T\}$ is determined by the stopped record. The physical law and the stopped law therefore assign it the same probability, written $P_\theta^\pi(A^\pi)$. We apply change of measure to the stopped laws and evaluate safety under the full physical laws.

For later use, define the one-step conditional divergence

$$\mathrm{kl}_t^{\theta,\theta'} := D_{\mathrm{KL}}(Q_{\theta,t}(\cdot \mid H_{t-1}, U_t) \,\|\, Q_{\theta',t}(\cdot \mid H_{t-1}, U_t))\,. \tag{17}$$

Definition 4 (Pairwise pre-commitment information)**.** The maximum information that can be collected safely before commitment is

$$I_{\mathrm{pre}}(T;\theta,\theta') := \sup_{\pi \in \Pi_{\mathrm{safe}}^T(\{\theta,\theta'\})} D_{\mathrm{KL}}\Big(P_{\theta,T}^{\pi,\mathrm{pre}} \,\Big\|\, P_{\theta',T}^{\pi,\mathrm{pre}}\Big)\,. \tag{18}$$

Its horizon-uniform capacity is

$$\beta_{\mathrm{pre}}(\theta,\theta') := \sup_{T\ge 1} I_{\mathrm{pre}}(T;\theta,\theta'). \tag{19}$$

Lemma 1 (Stopped controlled-KL identity)**.** *Assume $Q_{\theta,t}(\cdot \mid h,u) \ll Q_{\theta',t}(\cdot \mid h,u)$ for every pre-commitment history-action pair. Then, for every causal policy,*

$$D_{\mathrm{KL}}\Big(P_{\theta,T}^{\pi,\mathrm{pre}} \,\Big\|\, P_{\theta',T}^{\pi,\mathrm{pre}}\Big) = \mathbb{E}_\theta^\pi\left[\sum_{t=1}^T \mathbf{1}\{t < \tau^\pi\}\mathrm{kl}_t^{\theta,\theta'}\right]. \tag{20}$$

Proof. Since τ^π is an $\{\mathcal{F}_t\}$-stopping time with $\mathcal{F}_t = \sigma(H_{t-1}, U_t)$, the event $\{t < \tau^\pi\}$ is $\mathcal{F}_t$-measurable, hence determined by (H_{t-1}, U_t) before Z_t is seen. Form the likelihood ratio of the stopped record under θ against θ'. The policy kernels $\pi_t(\cdot \mid H_{t-1})$ are common to both environments and cancel, including the kernel that generates the committing action U_{τ^π}; only the observation kernels $Q_{\theta,t}$ versus $Q_{\theta',t}$ survive, and the stopping time contributes no additional likelihood factor.

If $\tau^\pi = m \le T$, the stopped record contains observations only through time $m-1$, so the log-likelihood ratio is the sum of one-step observation contributions for $t < m$; if no commitment occurs, the sum runs through $t = 1, \ldots, T$. In both cases the terms present are exactly those on $\{t < \tau^\pi\}$. Conditioning the t-th term on (H_{t-1}, U_t) gives $\mathrm{kl}_t^{\theta,\theta'}$ by definition (17), and taking expectations under θ yields (20). ☐

Finite-headroom information. After choosing u_t, the controller observes $y_t = \theta u_t + \xi_t$, where $\xi_t \sim \mathcal{N}(0,\sigma^2)$. The one-step divergence is

$$\mathrm{kl}_t^{\theta_b,\theta_d} = D_{\mathrm{KL}}\big(\mathcal{N}(\theta_b u_t, \sigma^2) \,\|\, \mathcal{N}(\theta_d u_t, \sigma^2)\big) = \frac{(\theta_d - \theta_b)^2 u_t^2}{2\sigma^2}. \tag{21}$$

Before commitment, $\sum_{t<\tau^\pi} u_t^2 \le B_d = d_0/\theta_d$. Hence Lemma 1 gives

$$I_{\mathrm{pre}}(T;\theta_b,\theta_d) \le \beta_{\mathrm{th}} := \frac{(\theta_d-\theta_b)^2 d_0}{2\sigma^2\theta_d}. \tag{22}$$

This bound does not depend on T.

Definition 5 (Commitment allowance)**.** For an ordered pair (θ, θ') with a nonempty uniformly safe class, define

$$\alpha_T(\theta, \theta') := \sup_{\pi \in \Pi^T_{\mathrm{safe}}(\{\theta, \theta'\})} P^{\pi}_{\theta'}(A^{\pi}). \tag{23}$$

We write α_T when the pair is clear. Any upper bound $\bar{\alpha}_T \geq \alpha_T$ can replace it in the bounds below. In particular, if

$$A^{\pi} \subseteq \mathsf{Safe}^c_{\theta',T} \quad \overline{P}^{\pi}_{\theta',T}\text{-a.s.}$$

for every uniformly safe policy, then $\alpha_T \leq \eta_T$.

Finite-headroom device. Crossing the θ_d-safe budget gives

$$d_{\tau^{\pi}+1} = d_0 - \theta_d \sum_{s \leq \tau^{\pi}} u_s^2 < 0$$

under θ_d. Hence commitment implies a safety violation and $\alpha_T(\theta_b, \theta_d) \leq \eta_T$.

3.2 The noncommitment regret profile

The stopped information bound gives a lower bound on the probability of not committing under the target. The next definition records the exact regret cost of that probability.

Definition 6 (Noncommitment regret profile)**.** For an ordered pair (θ, θ') and $q \in [0, 1]$, define

$$\mathcal{G}_T(\theta, \theta'; q) := \inf_{\substack{\mu \in \Pi^T_{\mathrm{safe}}(\{\theta, \theta'\}) \\ P^{\mu}_{\theta}((A^{\mu})^c) \geq q}} R_T(\mu; \theta), \tag{24}$$

with the convention $\inf \varnothing = +\infty$. Equivalently, whenever the constraint set is nonempty,

$$\mathcal{G}_T(\theta, \theta'; q) = J^{\star}_T(\theta) - \sup_{\substack{\mu \in \Pi^T_{\mathrm{safe}}(\{\theta, \theta'\}) \\ P^{\mu}_{\theta}((A^{\mu})^c) \geq q}} J_T(\mu; \theta). \tag{25}$$

The function $q \mapsto \mathcal{G}_T(\theta, \theta'; q)$ is nonnegative and nondecreasing.

This definition imposes no pathwise reward ceiling. In a stochastic-reward problem, the profile may be bounded directly in expectation. The following lemma gives a convenient pathwise bound for the deterministic examples.

Lemma 2 (Pathwise bound on the profile)**.** *Suppose there are constants*

$$0 \leq V_T(\theta, \theta') \leq B_T(\theta) \leq R_{\max,T} < \infty$$

such that every $\mu \in \Pi^T_{\mathrm{safe}}(\{\theta, \theta'\})$ *satisfies,* $\overline{P}^{\mu}_{\theta,T}$*-almost surely,*

$$\mathcal{R}_T \leq \begin{cases} V_T(\theta, \theta'), & \text{on } (A^{\mu})^c \cap \mathsf{Safe}_{\theta,T}, \\ B_T(\theta), & \text{on } A^{\mu} \cap \mathsf{Safe}_{\theta,T}, \\ R_{\max,T}, & \text{always.} \end{cases}$$

Then, for every $q \in [0, 1]$,

$$\begin{aligned} \mathcal{G}_T(\theta, \theta'; q) \geq \Big[& J^{\star}_T(\theta) - B_T(\theta) + \big(B_T(\theta) - V_T(\theta, \theta')\big) q \\ & - \big(R_{\max,T} - V_T(\theta, \theta')\big) \eta_T \Big]_+ . \end{aligned} \tag{26}$$

Proof. Fix a feasible policy μ in (24), and write

$$p := P_\theta^\mu((A^\mu)^c), \qquad f := \overline{P}_{\theta,T}^\mu(\mathsf{Safe}_{\theta,T}^c).$$

Let $x := \overline{P}_{\theta,T}^\mu((A^\mu)^c \cap \mathsf{Safe}_{\theta,T})$. Then $x \geq p - f$. The pathwise bounds give

$$\begin{aligned} J_T(\mu;\theta) &\leq V_T x + B_T(1-f-x) + R_{\max,T} f \\ &\leq B_T - (B_T - V_T)p + (R_{\max,T} - V_T)f \\ &\leq B_T - (B_T - V_T)q + (R_{\max,T} - V_T)\eta_T, \end{aligned}$$

where the arguments (θ, θ') have been suppressed. Subtracting from $J_T^\star(\theta)$ and taking the infimum over feasible policies gives (26). □

A useful simpler bound is

$$\mathcal{G}_T(\theta,\theta';q) \geq \left[G_T(\theta,\theta')q - \zeta_T(\theta,\theta')\right]_+, \qquad q \in [0,1]. \tag{27}$$

The quantities G_T and ζ_T are derived from the system; they are not assumptions of the main theorem.

Corollary 1 (Pathwise verification in deterministic systems)**.** *Suppose $J_{T,\mathrm{hard}}^\star(\theta)$ is achieved by a hard-safe policy and bounds the reward of every hard-safe trajectory. Suppose also that every safe noncommitting trajectory earns at most $V_{T,\mathrm{com}}(\theta,\theta')$ and that $0 \leq \mathcal{R}_T \leq R_{\max,T}$. Then*

$$\begin{aligned} \mathcal{G}_T(\theta,\theta';q) \geq \Big[&J_T^\star(\theta) - J_{T,\mathrm{hard}}^\star(\theta) + G_T(\theta,\theta')q \\ &- \big(R_{\max,T} - V_{T,\mathrm{com}}(\theta,\theta')\big)\eta_T \Big]_+, \end{aligned} \tag{28}$$

where

$$G_T(\theta,\theta') := J_{T,\mathrm{hard}}^\star(\theta) - V_{T,\mathrm{com}}(\theta,\theta'). \tag{29}$$

Since every hard-safe policy is chance-safe, $J_T^\star(\theta) \geq J_{T,\mathrm{hard}}^\star(\theta)$. Thus (27) *holds with this G_T and $\zeta_T = R_{\max,T}\eta_T$.*

These conditions hold in the deterministic examples in this paper. For stochastic rewards, the main theorem still applies; one bounds $\mathcal{G}_T(\theta,\theta';q)$ directly in expectation.

Remark 3 (Intuition)*.* The pathwise bounds involve four values, ordered as

$$V_{T,\mathrm{com}}(\theta,\theta') \;\leq\; J_{T,\mathrm{hard}}^\star(\theta) \;\leq\; J_T^\star(\theta) \;\leq\; R_{\max,T}.$$

$R_{\max,T}$ is the physical ceiling: the most any trajectory can earn, safe or not. $J_T^\star(\theta)$ is the benchmark: the best chance-η_T-safe value with θ known. $J_{T,\mathrm{hard}}^\star(\theta)$ is the best value with θ known and no violation tolerance; it enters for a pathwise reason: on $\mathsf{Safe}_{\theta,T}$ every realized trajectory is hard-safe, so in a deterministic-reward system its reward cannot exceed $J_{T,\mathrm{hard}}^\star(\theta)$, whereas the chance-safe value is an expectation and bounds no single path. Using $J_{T,\mathrm{hard}}^\star \leq J_T^\star$ only makes the regret bound conservative. $V_{T,\mathrm{com}}(\theta,\theta')$ is the recoverable value: the most a trajectory can earn under θ while retaining a hard-safe continuation under θ', that is, while noncommitting. The gap $G_T = J_{T,\mathrm{hard}}^\star - V_{T,\mathrm{com}}$ is the portion of the target's value reachable only by committing. The linear profile bound (27) reads: forcing noncommitment probability q forfeits at least a q-fraction of this gap. On the finite-headroom device, $R_{\max,T} = UT$ (full command every round), $J_{T,\mathrm{hard}}^\star(\theta_b) = \min\{UT, \sqrt{TB_b}\}$ (spend the true budget), $V_{T,\mathrm{com}} = \min\{UT, \sqrt{TB_d}\}$ (spend only the worn budget); G_T is the reward available only beyond the worn budget.

Finite-headroom device. For the target θ_b and alternative θ_d,

$$J^{\star}_{T,\mathrm{hard}}(\vartheta) = \min\{UT, \sqrt{TB_\vartheta}\}, \qquad B_0 = +\infty,$$

and every safe noncommitting trajectory earns at most

$$V_{T,\mathrm{com}}(\theta_b, \theta_d) = \min\{UT, \sqrt{TB_d}\}.$$

Thus

$$G_T(\theta_b, \theta_d) = \min\{UT, \sqrt{TB_b}\} - \min\{UT, \sqrt{TB_d}\}. \tag{30}$$

Since $R_{\max,T} = UT$, Corollary 1 gives

$$\mathcal{G}_T(\theta_b, \theta_d; q) \geq \left[G_T(\theta_b, \theta_d) q - UT\eta_T\right]_+ . \tag{31}$$

3.3 The binary limit

For $p, q \in [0, 1]$, let

$$d_{\mathrm{bin}}(p \| q) := p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q},$$

with the usual extended-value conventions. For $\alpha \in [0, 1)$ and $D \in [0, \infty]$, define

$$\psi_\alpha(D) := \inf \left\{p \in [0, 1-\alpha] : d_{\mathrm{bin}}(p \| 1-\alpha) \leq D\right\}. \tag{32}$$

Set $\psi_1(D) := 0$ and $\psi_\alpha(\infty) := 0$. The function ψ_α is nonincreasing. It is the smallest target noncommitment probability compatible with divergence D and alternative commitment probability at most α.

Theorem 1 (Binary pre-commitment limit). *Fix an ordered pair (θ, θ') whose uniformly safe class is nonempty. For $\pi \in \Pi^T_{\mathrm{safe}}(\{\theta, \theta'\})$, define*

$$D^\pi_T := D_{\mathrm{KL}}\left(P^{\pi,\mathrm{pre}}_{\theta,T} \,\middle\|\, P^{\pi,\mathrm{pre}}_{\theta',T}\right), \qquad a^\pi_T := P^\pi_{\theta'}(A^\pi).$$

Then

$$R_T(\pi; \theta) \geq \mathcal{G}_T\left(\theta, \theta'; \psi_{a^\pi_T}(D^\pi_T)\right). \tag{33}$$

Consequently,

$$R_T(\pi; \theta) \geq \mathcal{G}_T\left(\theta, \theta'; \psi_{\alpha_T(\theta,\theta')}(I_{\mathrm{pre}}(T; \theta, \theta'))\right). \tag{34}$$

Since

$$\psi_\alpha(D) \geq \left[\frac{1}{2} e^{-D} - \alpha\right]_+,$$

this implies the explicit bound

$$R_T(\pi; \theta) \geq \mathcal{G}_T\left(\theta, \theta'; \left[\frac{1}{2} e^{-I_{\mathrm{pre}}(T;\theta,\theta')} - \alpha_T(\theta, \theta')\right]_+\right). \tag{35}$$

If the linear profile bound (27) has been verified, then

$$R_T(\pi; \theta) \geq \left[G_T(\theta, \theta') \left(\frac{1}{2} e^{-I_{\mathrm{pre}}(T;\theta,\theta')} - \alpha_T(\theta, \theta')\right) - \zeta_T(\theta, \theta')\right]_+. \tag{36}$$

If, in addition, $\beta_{\mathrm{pre}}(\theta, \theta') < \infty$, $\alpha_T(\theta, \theta') \to 0$, and $\zeta_T(\theta, \theta') = o(G_T(\theta, \theta'))$, then

$$\liminf_{T\to\infty} \inf_{\pi \in \Pi^T_{\mathrm{safe}}(\{\theta,\theta'\})} \frac{\sup_{\vartheta \in \{\theta,\theta'\}} R_T(\pi; \vartheta)}{G_T(\theta, \theta')} \geq \frac{1}{2} e^{-\beta_{\mathrm{pre}}(\theta,\theta')}. \tag{37}$$

Interpretation. Equation 36 is the central reduction; the exponential expression in Equation 35 is a convenient relaxation. Safety controls the commitment probability under the alternative, the stopped experiment controls how different the target commitment probability can be, and the noncommitment profile converts the remaining target-side probability into regret.

Proof. Let

$$p := P_\theta^\pi((A^\pi)^c), \qquad q := P_{\theta'}^\pi((A^\pi)^c) = 1 - a_T^\pi.$$

Data processing for the event $(A^\pi)^c$ gives

$$d_{\mathrm{bin}}(p\|q) \le D_T^\pi. \tag{38}$$

If $p \le q$, the definition of ψ gives $p \ge \psi_{a_T^\pi}(D_T^\pi)$; if $p > q$, the same inequality is immediate. The policy π is therefore feasible in the definition of the profile at this value, which proves (33).

For the uniform bound, Definition 5 gives $q \ge 1-\alpha_T(\theta,\theta')$. If $p \le 1-\alpha_T$, then $d_{\mathrm{bin}}(p\|1-\alpha_T) \le d_{\mathrm{bin}}(p\|q)$; if $p > 1-\alpha_T$, the same lower bound is immediate. Hence

$$p \ge \psi_{\alpha_T(\theta,\theta')}(D_T^\pi).$$

Since

$$D_T^\pi \le I_{\mathrm{pre}}(T;\theta,\theta')$$

and ψ_α is nonincreasing, (34) follows. Bretagnolle–Huber gives

$$\psi_\alpha(D) \ge \left[\frac{1}{2}e^{-D} - \alpha\right]_+.$$

This yields (35). Substituting (27) gives (36). The asymptotic statement follows from the horizon-uniform information bound and the stated limits. □

The explicit term in (35) is positive when

$$I_{\mathrm{pre}}(T;\theta,\theta') < \log\frac{1}{2\alpha_T(\theta,\theta')}.$$

For $\alpha_T = T^{-2}$, the threshold is $2\log T - \log 2$.

Finite-headroom device. Using $\alpha_T(\theta_b,\theta_d) \le \eta_T$, (22), and (31), Theorem 1 gives

$$R_T(\pi;\theta_b) \ge \left[G_T(\theta_b,\theta_d)\left(\frac{1}{2}e^{-\beta_{\mathrm{th}}} - \eta_T\right) - UT\eta_T\right]_+. \tag{39}$$

With $\eta_T = T^{-2}$, fixed $0 < \theta_b < \theta_d$, and an inactive per-step cap, $G_T(\theta_b,\theta_d) = \Theta(\sqrt{T})$. At $\theta_b = 0$, $G_T(0,\theta_d) = UT - O(\sqrt{T}) = \Theta(T)$, so the same information bound gives linear regret.

Corollary 2 (Information required for a target regret)**.** *Suppose the linear profile bound* (27) *holds. If a uniformly safe policy satisfies $R_T(\pi;\theta) \le r_T$, then*

$$I_{\mathrm{pre}}(T;\theta,\theta') \ge \left[\log\frac{1}{2\left(\alpha_T(\theta,\theta') + \frac{r_T+\zeta_T(\theta,\theta')}{G_T(\theta,\theta')}\right)}\right]_+. \tag{40}$$

Thus $r_T = o(G_T)$ requires unbounded pre-commitment information whenever $\alpha_T \to 0$ and $\zeta_T = o(G_T)$.

Proof. Apply (36) and rearrange. □

Remark 4 (Other risk constraints)*.* The theorem uses safety only through the commitment allowance α_T. Any risk constraint that bounds $P^\pi_{\theta'}(A^\pi)$ can be used. For example, if a nonnegative loss satisfies $L_T \geq m_T\mathbf{1}\{A^\pi\}$ under θ' and $\mathrm{CVaR}_\beta(L_T) \leq c_T$, then

$$P^\pi_{\theta'}(A^\pi) \leq \frac{c_T}{m_T}.$$

Specifications that budget cumulative violations rather than bounding the violation probability under every plausible model need not imply a small commitment allowance and therefore fall outside the premise of the theorem; in the safe linear bandit setting, [9] show that such a budget purchases exactly the boundary exploration that uniform chance safety forbids, and near-optimal learning is then possible.

Remark 5 (Generality)*.* The theorem uses only the two stopped laws and data processing. It does not require Markov dynamics, Gaussian noise, linearity, full-state observation, or absolute continuity. Absolute continuity is needed only for the sum identity in Lemma 1. An application must define a predictable commitment rule and bound α_T, I_{pre}, and $\mathcal{G}_T$.

3.4 Continuous classes: optimizing over the alternative

The binary theorem applies to every incompatible pair in Θ_0. If Θ_0 is continuous, the pairwise bound can be maximized over the alternative.

Corollary 3 (Worst-case alternative and minimax bound)**.** *Let Θ_0 be arbitrary. For a target $\theta \in \Theta_0$, let $\mathcal{A}(\theta) \subseteq \Theta_0$ contain alternatives with predictable commitment rules. Each alternative may use its own rule. Then every $\pi \in \Pi^T_{\mathrm{safe}}(\Theta_0)$ satisfies*

$$R_T(\pi;\theta) \geq \sup_{\theta' \in \mathcal{A}(\theta)} \mathcal{G}_T\big(\theta,\theta';\psi_{\alpha_T(\theta,\theta')}(I_{\mathrm{pre}}(T;\theta,\theta'))\big)\,, \tag{41}$$

and

$$\mathfrak{R}_T(\Theta_0) \geq \sup_{\theta\in\Theta_0} \sup_{\theta' \in \mathcal{A}(\theta)} \mathcal{G}_T\big(\theta,\theta';\psi_{\alpha_T(\theta,\theta')}(I_{\mathrm{pre}}(T;\theta,\theta'))\big)\,. \tag{42}$$

Proof. Fix $\theta' \in \mathcal{A}(\theta)$. A policy safe on Θ_0 is safe on $\{\theta,\theta'\}$, so Theorem 1 applies. Taking the supremum over alternatives gives (41). Taking the supremum over targets and then the infimum over uniformly safe policies gives (42). □

If each pair satisfies the linear profile bound (27), Corollary 3 gives the explicit lower bound

$$R_T(\pi;\theta) \geq \sup_{\theta'\in\mathcal{A}(\theta)} \left[G_T(\theta,\theta')\left(\frac{1}{2}e^{-I_{\mathrm{pre}}(T;\theta,\theta')} - \alpha_T(\theta,\theta')\right) - \zeta_T(\theta,\theta')\right]_+. \tag{43}$$

The optimization balances value and distinguishability. Nearby alternatives are harder to distinguish but may have a smaller value gap.

The finite-headroom device on a continuum. Let $\Theta_0 = [0,\bar\theta]$, with $B_\vartheta = d_0/\vartheta$ for $\vartheta > 0$ and $B_0 = +\infty$. For a target θ and alternative $\theta' \in (\theta,\bar\theta]$, commitment is the first exit from the θ'-budget. Then

$$\alpha_T(\theta,\theta') \leq \eta_T, \qquad \zeta_T(\theta,\theta') = UT\eta_T,$$

$$G_T(\theta,\theta') = \min\{UT,\sqrt{TB_\theta}\} - \min\{UT,\sqrt{TB_{\theta'}}\},$$

and

$$I_{\mathrm{pre}}(T;\theta,\theta') \leq \beta(\theta,\theta') := \frac{(\theta'-\theta)^2 d_0}{2\sigma^2\theta'}.$$

Ignoring the vanishing η_T terms, define

$$S_T(\theta,\theta') := \frac{1}{2}e^{-\beta(\theta,\theta')}G_T(\theta,\theta').$$

Interior target. Let $\theta > 0$ and suppose the per-step cap is inactive. Then

$$S_T(\theta,\theta') = \frac{1}{2}\sqrt{Td_0}\, f_\theta(\theta'), \qquad f_\theta(\theta') := e^{-\beta(\theta,\theta')}\big(\theta^{-1/2} - \theta'^{-1/2}\big).$$

The function f_θ does not depend on T, so the optimized lower bound is $\Theta(\sqrt{T})$.

Boundary target. For $\theta = 0$,

$$G_T(0,\theta') = UT - \sqrt{Td_0/\theta'}, \qquad \beta(0,\theta') = \frac{\theta' d_0}{2\sigma^2}.$$

Optimizing $S_T(0,\theta')$ gives

$$\theta'^\star = \left(\frac{\sigma^4}{U^2 d_0}\right)^{1/3} T^{-1/3}(1+o(1)), \qquad \beta^\star = \frac{1}{2}\left(\frac{d_0}{\sigma U}\right)^{2/3} T^{-1/3}(1+o(1)), \tag{44}$$

and

$$\sup_{\theta'\in(0,\bar\theta]} S_T(0,\theta') = \frac{UT}{2} - \frac{3}{4}\left(\frac{d_0^2 U}{\sigma^2}\right)^{1/3} T^{2/3} + o(T^{2/3}). \tag{45}$$

Thus

$$\mathfrak{R}_T([0,\bar\theta]) \geq \frac{UT}{2}(1-o(1)). \tag{46}$$

The optimizing alternative approaches the boundary target at rate $T^{-1/3}$.

3.5 One policy, many alternatives: safe lists

Remark 6 (One policy against many alternatives). The binary theorem is pairwise. Its information profile may use a different policy for each alternative. If the learner must choose among M model-specific regimes, one policy must gather information about all M possibilities. The relevant quantity is

$$I^{(M)}_{\text{pre}}(T;\Theta_M) := \sup_{\pi\in\Pi^T_{\text{safe}}(\Theta_M)} I_\pi(J; S^\pi_T), \qquad J \sim \mathrm{Unif}[M]. \tag{47}$$

When partial identification already improves control, the target is a certified candidate list rather than exact classification.

Proposition 1 (Information required for a safe candidate list)**.** *Let $J \sim \mathrm{Unif}[M]$, let S be the information available when a nonempty list $C(S) \subseteq [M]$ is selected, and fix $L \in \{1,\dots,M-1\}$. Define*

$$\varepsilon_L := P\big(J \notin C(S) \text{ or } |C(S)| > L\big). \tag{48}$$

Then

$$I(J;S) \geq (1-\varepsilon_L)\log\frac{M}{L} - h_2(\varepsilon_L), \tag{49}$$

where $h_2(p) := -p\log p - (1-p)\log(1-p)$. Consequently, if

$$P(J \notin C(S)) \leq \eta, \qquad P(|C(S)| > L) \leq \delta, \qquad \eta+\delta \leq \tfrac{1}{2},$$

then

$$I(J;S) \geq (1-\eta-\delta)\log\frac{M}{L} - h_2(\eta+\delta). \tag{50}$$

The proof is in the appendix. The case $L = M$ gives no reduction, $L = 1$ gives exact specialization, and $1 < L < M$ gives partial specialization.

Remark 7 (Scalar continuous candidate sets). Let $J \sim \mathrm{Unif}[\theta_{\min}, \theta_{\max}]$, with $D = \theta_{\max} - \theta_{\min}$. Suppose the learner returns a set $C(S)$ such that

$$P\big(J \notin C(S) \text{ or } \operatorname{diam} C(S) > \ell\big) \le \varepsilon \le \tfrac{1}{2}.$$

Then

$$I(J; S) \ge (1-\varepsilon)\log\frac{D}{\ell} - h_2(\varepsilon), \qquad \ell \ge D\exp\left(-\frac{I(J;S) + h_2(\varepsilon)}{1-\varepsilon}\right). \tag{51}$$

This is a converse for a scalar parameter on an interval. Achieving the bound still requires a safe sensing policy.

3.6 How to use the limits in an application

For a new system:

1. define a predictable commitment event;
2. bound the commitment allowance α_T under the incompatible alternative;
3. compute or lower-bound the noncommitment regret profile $\mathcal{G}_T$; and
4. bound the stopped KL divergence, or the stopped mutual information in a multi-alternative problem.

The theorem then gives a regret lower bound. A positive result requires a single uniformly safe policy that gathers the needed information and moves safely to the selected regime.

4 Mechanisms and Examples

Section 3.6 gives the application procedure for the lower bound. Here we study two mechanisms that bound the information available before commitment: finite expenditure and a finite sensing window. We apply them to a constrained linear-regulation problem and an unknown-location problem. The first yields linear regret; the second yields safe partial adaptation through a certified candidate list.

4.1 Finite expenditure before commitment

Lemma 3 (Finite pre-commitment expenditure)**.** *Fix an ordered pair (θ, θ') and its commitment time τ^π. Suppose there is a nonnegative predictable quantity $S_t = S(H_{t-1}, U_t)$, constants $\kappa < \infty$ and $q \ge 1$, and a number $B_T < \infty$ such that every $\pi \in \Pi^T_{\mathrm{safe}}(\{\theta, \theta'\})$ satisfies, pathwise,*

$$\sum_{t=1}^{T} \mathbf{1}\{t < \tau^\pi\} S_t \le B_T, \tag{52}$$

and, whenever $t < \tau^\pi$,

$$\mathrm{kl}_t^{\theta,\theta'} \le \kappa S_t^q. \tag{53}$$

Then

$$I_{\mathrm{pre}}(T; \theta, \theta') \le \kappa B_T^q. \tag{54}$$

In particular, if $\sup_T B_T < \infty$, then $\beta_{\mathrm{pre}}(\theta, \theta') < \infty$.

The proof is in Appendix A.1. Since $\sum_{t<\tau^\pi} S_t^q \le (\sum_{t<\tau^\pi} S_t)^q$ for $q \ge 1$, the pathwise expenditure bound directly bounds the stopped information. For $q = 1$, information is at most proportional to expenditure. For $q > 1$, splitting a finite budget among many small actions cannot produce persistent information.

Remark 8 (One-sided versus Signed Actions). The conversion

$$\sum_t u_t \le B \quad \Longrightarrow \quad \sum_t u_t^2 \le \min\{B^2, UB\}$$

requires $u_t \ge 0$. With signed inputs and only a constraint on the signed sum, cancellations can keep $\sum_t u_t$ bounded while $\sum_t u_t^2$ grows. This caveat is specific to the linear one-sided margin of Variant I. In Variant II, safety directly bounds the signed expenditure $\sum_t (a^\top u_t)^2$, so reversing the active loading consumes the same quadratic headroom rather than restoring it. The information ceiling therefore persists under fully signed requests and controls.

4.2 Constrained linear regulation under two safety geometries

We use a common two-channel output geometry to exhibit two finite-expenditure mechanisms in linear–quadratic regulation. Variant I uses nonnegative reserve deployment and a linear security margin; its information ceiling follows from an order mismatch between linear expenditure and quadratic Gaussian information. Variant II permits genuinely signed regulation and uses a quadratic I^2t-type margin. The second variant shows that the obstruction is not an artifact of one-sided actuation.

Shared observation geometry. Let

$$a \in \{a_0, a_1\}, \qquad a_0 = e_2, \qquad a_1 = e_1, \tag{55}$$

and write $u_t = (p_t, q_t)$. The regulated state x_t is observed before the action. After applying u_t, the controller observes

$$y_t = a^\top u_t + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \tag{56}$$

The active direction and the physical margin are not directly observed. The initial margin and its update law are known, so the controller can propagate the candidate margin associated with each direction from its action history. In both variants the target is a_0: the p-channel is then invisible to the active security constraint, whereas under a_1 that same channel spends the margin.

In each variant the controller minimizes a quadratic cost and is compared with the parameter-aware chance-safe oracle. Stable dynamics and bounded inputs give a uniform stage-cost ceiling. Subtracting the stage cost from this ceiling converts the example to the bounded-reward convention without changing regret, with an $O(T\eta_T)$ value-profile remainder.

Variant I: one-sided reserve deployment. Let

$$u_t = (p_t, q_t) \in [0, U]^2, \qquad x_{t+1} = \rho x_t + w - p_t, \qquad x_1 = \bar{x} := \frac{w}{1-\rho}, \tag{57}$$

where $0 < \rho < 1$ and $0 < w \le U$, and define

$$C_T^{(1)}(\pi; a) := \mathbb{E}_a^\pi \sum_{t=1}^T \left[x_{t+1}^2 + \lambda(p_t^2 + q_t^2) \right]. \tag{58}$$

Let $C_T^{(1),\star}(a)$ be the parameter-aware chance-safe oracle cost and write

$$R_T^{(1)}(\pi; a) := C_T^{(1)}(\pi; a) - C_T^{(1),\star}(a).$$

The hidden security margin evolves as

$$d_{t+1} = d_t - a^\top u_t, \qquad d_1 = d_0 > 0, \qquad d_t \ge 0. \tag{59}$$

Under a_0, the feasible policy $p_t = w$, $q_t = 0$ gives

$$C_T^{(1),\star}(a_0) \le \lambda w^2 T + O(1). \tag{60}$$

Define commitment relative to a_1 by

$$\tau_1^\pi := \inf\left\{ t \le T : \sum_{s=1}^t p_s > d_0 \right\}, \qquad A_1^\pi := \{\tau_1^\pi \le T\}. \tag{61}$$

Commitment violates safety under a_1, so $\mathbb{P}_{a_1}^\pi(A_1^\pi) \le \eta_T$. On noncommitment, $\sum_t p_t \le d_0$. Since

$$x_{t+1} = \bar{x} - \sum_{s=1}^t \rho^{t-s} p_s,$$

the deviations from $\bar{x}$ sum to at most $d_0/(1-\rho)$, and every noncommitting trajectory satisfies

$$\sum_{t=1}^T x_{t+1}^2 \ge \bar{x}^2 T - \frac{2\bar{x} d_0}{1-\rho}.$$

Thus the common-regime gap can be chosen so that

$$G_T^{(1)} \ge (\bar{x}^2 - \lambda w^2)T - O(1), \tag{62}$$

and hence $G_T^{(1)} = \Theta(T)$ whenever $\lambda w^2 < \bar{x}^2$. The input box also gives

$$|x_t| \le \max\left\{ \bar{x}, \frac{U-w}{1-\rho} \right\},$$

which verifies the bounded stage-cost claim above.

Under a_0 and a_1, the residual means are q_t and p_t, respectively, so

$$\mathrm{kl}_t(a_0, a_1) = \frac{(q_t - p_t)^2}{2\sigma^2}. \tag{63}$$

Before commitment, $\sum p_t \le d_0$ pathwise. On the target-safe event, $\sum q_t \le d_0$; on its complement, which has probability at most η_T, only the input box remains. Splitting the stopped-KL expectation over these events gives

$$I_{\mathrm{pre}}^{(1)}(T; a_0, a_1) \le \frac{\min\{4d_0^2, 2Ud_0\}}{2\sigma^2} + \frac{TU^2\eta_T}{2\sigma^2}. \tag{64}$$

For $\eta_T \le T^{-2}$, the profile is uniformly bounded, and Theorem 1 gives

$$R_T^{(1)}(\pi; a_0) = \Omega(T).$$

Continuous cross-loading. The same one-sided model also yields a useful continuous-class calculation. Replace the binary loading class by

$$a_\gamma = (\gamma, 1)^\top, \qquad \gamma \in [0, \bar{\gamma}], \tag{65}$$

with target $\gamma = 0$, so that

$$d_{t+1} = d_t - q_t - \gamma p_t, \qquad y_t = q_t + \gamma p_t + \xi_t. \tag{66}$$

For an alternative $\gamma > 0$, let commitment be the first time $\sum_{s \le t}(q_s + \gamma p_s) > d_0$. This is a safety violation under γ, so the commitment allowance is at most η_T. Strictly before commitment,

$$\sum_t p_t \le \frac{d_0}{\gamma}.$$

The q-channel cancels from the likelihood ratio, and the same budget also bounds the target value on noncommitment. Therefore

$$I_{\mathrm{pre}}(T;0,\gamma) \le \frac{\gamma U d_0}{2\sigma^2}, \qquad G_T^{(1)}(\gamma) \ge cT - \frac{B}{\gamma} - O(1), \tag{67}$$

where

$$c := \bar{x}^2 - \lambda w^2 > 0, \qquad B := \frac{2\bar{x} d_0}{1-\rho}.$$

Choose the horizon-dependent alternative $\gamma_T = T^{-1/2}$. Then $I_{\mathrm{pre}}(T;0,\gamma_T) = O(T^{-1/2})$ and $G_T^{(1)}(\gamma_T) \ge cT - O(\sqrt{T})$. For $\eta_T = T^{-2}$,

$$\psi_{\eta_T}(I_{\mathrm{pre}}(T;0,\gamma_T)) = 1 - o(1),$$

so Corollary 3 gives

$$R_T^{(1)}(\pi;0) \ge cT - o(T). \tag{68}$$

Thus even a cross-loading that vanishes as $T^{-1/2}$ can block asymptotically the entire linear regulation advantage.

Variant II: signed regulation with quadratic headroom. Let $s_t \in \{-1,+1\}$ be an exogenous up/down regulation request, observed before the action, included in the pre-action history, and independent of a. Take

$$u_t = (p_t, q_t) \in [-U,U]^2, \qquad x_{t+1} = \rho x_t + w s_t - p_t - q_t, \qquad x_1 = 0, \tag{69}$$

where $0 \le \rho < 1$ and $0 < w \le U$, with cost

$$C_T^{(2)}(\pi;a) := \mathbb{E}_a^\pi \sum_{t=1}^T \left[x_{t+1}^2 + \lambda(p_t^2 + q_t^2)\right]. \tag{70}$$

Let $C_T^{(2),\star}(a)$ be the parameter-aware chance-safe oracle cost and write

$$R_T^{(2)}(\pi;a) := C_T^{(2)}(\pi;a) - C_T^{(2),\star}(a).$$

The hidden margin now evolves as

$$d_{t+1} = d_t - (a^\top u_t)^2, \qquad d_1 = d_0 > 0, \qquad d_t \ge 0. \tag{71}$$

Under a_0, the feasible policy $p_t = w s_t$, $q_t = 0$ keeps $x_t = 0$ and certifies $C_T^{(2),\star}(a_0) \le \lambda w^2 T$. Define

$$\tau_2^\pi := \inf\left\{t \le T : \sum_{s=1}^t p_s^2 > d_0\right\}. \tag{72}$$

Commitment violates safety under a_1, so $\mathbb{P}_{a_1}^\pi(\tau_2^\pi \le T) \le \eta_T$.

On a target-safe, noncommitting trajectory,

$$\|p_{1:T} + q_{1:T}\|_2 \le 2\sqrt{d_0}.$$

With

$$e_t := w s_t - p_t - q_t = x_{t+1} - \rho x_t,$$

we obtain

$$\|e_{1:T}\|_2 \ge \left(w\sqrt{T} - 2\sqrt{d_0}\right)_+, \qquad \|e_{1:T}\|_2 \le (1+\rho)\|x_{2:T+1}\|_2.$$

Therefore the common-regime gap satisfies

$$G_T^{(2)} \geq \left[\frac{\left(w\sqrt{T} - 2\sqrt{d_0}\right)_+^2}{(1+\rho)^2} - \lambda w^2 T \right]_+ . \tag{73}$$

If $\lambda < (1+\rho)^{-2}$, then $G_T^{(2)} = \Theta(T)$. Moreover,

$$|x_t| \leq \frac{w + 2U}{1 - \rho},$$

so the stage cost is uniformly bounded here as well.

The one-step divergence remains

$$\mathrm{kl}_t(a_0, a_1) = \frac{(q_t - p_t)^2}{2\sigma^2}.$$

Before τ_2^π, $\sum p_t^2 \leq d_0$ pathwise. On the target-safe event, $\sum q_t^2 \leq d_0$; on its complement, the input box gives $\sum (q_t - p_t)^2 \leq 4U^2 T$. Hence

$$I_{\mathrm{pre}}^{(2)}(T; a_0, a_1) \leq \frac{2d_0}{\sigma^2} + \frac{2U^2 T \eta_T}{\sigma^2}. \tag{74}$$

Corollary 4 (Linear regret with an unknown active constraint direction)**.** *For $i \in \{1, 2\}$, let $\Pi_{\mathrm{safe}}^{T,i}$ denote the policies that are uniformly chance-η_T safe for $\{a_0, a_1\}$ in Variant i. Suppose that all system parameters are fixed independently of T and that $\eta_T \leq T^{-2}$.*

For Variant I, assume $g_1 := \bar{x}^{\,2} - \lambda w^2 > 0$, $\beta_1 := \frac{\min\{4d_0^2, 2Ud_0\}}{2\sigma^2}$. For Variant II, assume $g_2 := w^2\big((1+\rho)^{-2} - \lambda\big) > 0$, $\beta_2 := \frac{2d_0}{\sigma^2}$. Then, for $i \in \{1, 2\}$,

$$\liminf_{T\to\infty} \inf_{\pi \in \Pi_{\mathrm{safe}}^{T,i}} \frac{R_T^{(i)}(\pi; a_0)}{T} \geq \frac{g_i}{2} e^{-\beta_i} > 0.$$

In particular, there are constants $c_i > 0$ and $T_i < \infty$, depending only on the fixed system parameters, such that every $\pi \in \Pi_{\mathrm{safe}}^{T,i}$ satisfies

$$R_T^{(i)}(\pi; a_0) \geq c_i T, \qquad T \geq T_i.$$

Thus both safety geometries force linear oracle regret under uniform chance safety.

Proof. For Variant I, the noncommitment calculation above gives

$$G_T^{(1)} \geq g_1 T - O(1),$$

while the stopped-information bound gives

$$I_{\mathrm{pre}}^{(1)}(T; a_0, a_1) \leq \beta_1 + o(1).$$

For Variant II,

$$G_T^{(2)} \geq g_2 T - O(\sqrt{T}), \qquad I_{\mathrm{pre}}^{(2)}(T; a_0, a_1) \leq \beta_2 + o(1).$$

In both variants, commitment under a_1 implies a safety violation, so the commitment allowance is at most $\eta_T = o(1)$. The uniform stage-cost bounds give a noncommitment-profile remainder of order $O(T\eta_T) = o(T)$. Applying Theorem 1, dividing by T, and taking the lower limit proves

$$\liminf_{T\to\infty} \inf_{\pi \in \Pi_{\mathrm{safe}}^{T,i}} \frac{R_T^{(i)}(\pi; a_0)}{T} \geq \frac{g_i}{2} e^{-\beta_i}.$$

The eventual linear lower bound follows because the right-hand side is strictly positive. $\square$

Role of signed actuation. Variant I isolates the one-sided order mismatch and supplies the local continuous cross-loading calculation. Variant II shows that neither the information ceiling nor the linear-regret conclusion depends on one-sided actuation: the request and both reserve injections may change sign, while the active loading spends quadratic energy. Sign reversal can undo a linear signed expenditure, but it cannot undo an even expenditure: reversing the active loading pays the quadratic headroom a second time. In both variants, full observation of the regulated state is insufficient because the active security channel is visible only through the input-dependent residual. A persistent direct measurement of the margin would change the stopped information profile and can remove the obstruction.

4.3 Finite sensing bandwidth before an evolving deadline

The linear-regulation example limits information because informative control spends a hidden safety margin. A different obstruction occurs when the state evolves toward a boundary while the learner must search across many possible sensing locations.

Lemma 4 (Finite pre-commitment sensing window)**.** *Consider a labeled problem with $J \sim \mathrm{Unif}[M]$. Suppose that, under every policy, at most N_T label-dependent observations can occur before specialization or an absorbing fallback, and each such observation satisfies*

$$I_\pi(J; Z_t \mid H_{t-1}, U_t, t < \tau^\pi) \le c_{\mathrm{obs}}. \tag{75}$$

Then

$$I_{\mathrm{pre}}^{(M)}(T; \Theta_M) \le N_T c_{\mathrm{obs}}. \tag{76}$$

For the one-location Gaussian sensor

$$Z_t = \mu \mathbf{1}\{A_t = J\} + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, \sigma^2), \tag{77}$$

where $A_t \in [M]$ is chosen adaptively, one may take

$$c_{\mathrm{obs}} = \min\left\{\log 2, \frac{\mu^2}{2\sigma^2}\right\}. \tag{78}$$

The proof is in Appendix A.2. This is the one-policy search effect identified in Remark 6: one policy must search over all labels, while each round can inspect only one location. The same bandit-feedback search structure governs quickest changepoint detection with adaptive sensing [10], where the delay–false-alarm tradeoff arises.

4.4 Unknown disturbance location: safe lists and partial adaptation

There are M components and one unknown affected location $J \in [M]$. Fix an integer $1 \le N < T$. During diagnostic operation,

$$X_{i,t+1} = X_{i,t} + \Delta \mathbf{1}\{i = J\}, \qquad X_{i,1} = 0, \qquad i \in [M], \tag{79}$$

and safety requires $X_{i,t} \le H$ for all i, t. Write

$$H = N\Delta. \tag{80}$$

Thus full diagnostic operation is safe for N rounds; on the next round the affected component would cross the boundary unless mitigation is selected. At each diagnostic round the learner chooses one sensor $A_t \in [M]$ and observes

$$Z_t = \mu \mathbf{1}\{A_t = J\} + \xi_t, \qquad \xi_t \overset{\mathrm{iid}}{\sim} \mathcal{N}(0, \sigma^2). \tag{81}$$

Diagnostic operation earns unit reward per round. At the deadline the learner selects a nonempty certified list $\mathcal{C}_N \subseteq [M]$. The post-diagnostic controller divides both throughput and mitigation across the retained candidates:

$$q(\mathcal{C}_N) = \frac{1}{|\mathcal{C}_N|}, \qquad u_i(\mathcal{C}_N) = \frac{\Delta}{|\mathcal{C}_N|} \mathbf{1}\{i \in \mathcal{C}_N\}. \tag{82}$$

After the list is chosen,

$$X_{i,t+1} = X_{i,t} + \Delta q(\mathcal{C}_N)\mathbf{1}\{i = J\} - u_i(\mathcal{C}_N). \tag{83}$$

If $J \in \mathcal{C}_N$, the affected state is arrested. If $J \notin \mathcal{C}_N$, it violates the constraint on the next round; we treat this as an absorbing zero-reward failure. Thus safety of the list policy is equivalent to retaining the true location.

Proposition 1 and Lemma 4 give the following list-size bound. If a chance-safe policy returns a list $\mathcal{C}_N$ with $\mathbb{P}(J \notin \mathcal{C}_N) \le \eta$ and $\mathbb{P}(|\mathcal{C}_N| > L) \le \delta$, then necessarily

$$N\, c_{\text{sens}} \ge (1 - \eta - \delta) \log \frac{M}{L} - h_2(\eta + \delta), \qquad c_{\text{sens}} := \min\left\{\log 2, \frac{\mu^2}{2\sigma^2}\right\}. \tag{84}$$

Equivalently, for a deterministic size cap and failure probability at most $\eta \le 1/2$,

$$L \ge M \exp\left\{-\frac{N c_{\text{sens}} + h_2(\eta)}{1 - \eta}\right\}. \tag{85}$$

Finite pre-commitment information can therefore reduce the uncertainty set even when it is insufficient for exact localization.

One uniformly safe elimination rule is the following. For candidate j, let

$$\Lambda_{j,t} := \sum_{s \le t : A_s = j} \log \frac{\phi_{0,\sigma}(Z_s)}{\phi_{\mu,\sigma}(Z_s)}, \tag{86}$$

where $\phi_{m,\sigma}$ is the density of $\mathcal{N}(m, \sigma^2)$, and set

$$\mathcal{C}_t := \left\{ j : \Lambda_{j,t} < \log \frac{1}{\eta_T} \right\}. \tag{87}$$

Under the true location, $\exp(\Lambda_{J,t})$ is a likelihood-ratio martingale, so Ville's inequality gives

$$\mathbb{P}_J\big(J \notin \mathcal{C}_t \text{ for some } t \le N\big) \le \eta_T. \tag{88}$$

If the set in (87) is empty, the learner returns $[M]$. Otherwise, it can inspect, for example, the least-sampled surviving location. This rule is not claimed to be optimal; it only shows that partial certification can be uniformly safe.

Under the reward convention above, the robust, list-adaptive, and oracle values are

$$J_T^{\text{rob}} = N + \frac{T - N}{M}, \tag{89}$$

$$J_T^{\text{list}} = N + (T - N)\mathbb{E}\left[\frac{\mathbf{1}\{J \in \mathcal{C}_N\}}{|\mathcal{C}_N|}\right], \tag{90}$$

$$J_T^{\text{oracle}} = T. \tag{91}$$

The list policy improves on robust control when

$$\mathbb{E}\left[\frac{\mathbf{1}\{J \in \mathcal{C}_N\}}{|\mathcal{C}_N|}\right] > \frac{1}{M},$$

and it reaches the oracle only when $\mathcal{C}_N = \{J\}$ almost surely. Whenever the first inequality is strict and exact localization does not hold,

$$J_T^{\text{rob}} < J_T^{\text{list}} < J_T^{\text{oracle}}. \tag{92}$$

Section 7.3 evaluates this rule over a range of sensing windows.

Example	Pre-commitment information	Oracle gap	Consequence
Unknown active constraint direction (linear regulation)	bounded by finite margin expenditure.	$\Theta(T)$	linear regret under uniform safety
Unknown disturbance location	at most Nc_{sens} before mitigation	list dependent	certified robust, partial, or exact adaptation
Repeatable safe experiment	can grow linearly	model dependent	oracle recovery when the transition remains safe

Table 1: The effect of bounded information depends on the oracle gap. In a multi-alternative problem, finite information can instead determine the size of a certified safe list.

5 Safe Oracle Recovery from Repeatable Experiments

Theorem 1 gives the lower-bound certificate, and Section 4 verifies it in several systems. We now give a sufficient condition for recovery. If a safety-preserving experiment can be repeated and remains informative, a confidence-based controller can learn without leaving the safe set. The result uses three properties:
1. a conservative experiment is safe and returns the system to a common continuation state;
2. repeated experiments accumulate information about the unknown parameter; and
3. the loss in oracle value is controlled by the parameter error.

This is not a converse to the lower bound. It identifies one setting in which pre-commitment information is persistent and can be used safely.

5.1 Repeatable safe experiments

Let

$$\theta \in [\underline{\theta}, \overline{\theta}], \qquad 0 < \underline{\theta} < \overline{\theta},$$

be a fixed unknown scalar parameter. Time is divided into episodes. Let $\mathcal{G}_{t-1}$ denote the σ-field generated by the first $t-1$ episodes. The learner chooses a $\mathcal{G}_{t-1}$-measurable conservative design parameter ϑ_t and executes an experiment κ_{ϑ_t}. The experiment returns reward r_t and an observation

$$y_t = \theta a_t + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \tag{93}$$

where a_t is $\mathcal{G}_{t-1}$-measurable and ξ_t is independent of $\mathcal{G}_{t-1}$. Define the accumulated design information

$$V_t := \sum_{s=1}^{t} \frac{a_s^2}{\sigma^2}, \qquad V_0 := 0. \tag{94}$$

Let $v^\star(\theta) \in [0, r_{\max}]$ be the oracle value per episode. For $\vartheta \le \theta$, let $v_{\mathrm{bnd}}(\theta, \vartheta)$ be the expected value of κ_ϑ under the true parameter θ.

Assumption 1 (Repeatable safe experiments)**.** There is a nondecreasing function $\omega : [0, \infty) \to [0, r_{\max}]$ with $\omega(0) = 0$ such that:

(P1) *Fixed parameter.* The experiments do not change θ, and the same parameter governs all episodes.

(P2) *Safe repeatability.* For every $\underline{\theta} \le \vartheta \le \theta \le \overline{\theta}$, executing κ_ϑ is safe under θ and, possibly after a fixed reset, returns the system to a common state from which the next experiment can be run safely.

(P3) *Value modulus.* Whenever $\vartheta \le \theta$,

$$0 \le v^\star(\theta) - v_{\mathrm{bnd}}(\theta, \vartheta) \le \omega(\theta - \vartheta). \tag{95}$$

(P4) *Bounded reward.* Almost surely, $0 \le r_t \le r_{\max}$, and

$$\mathbb{E}_\theta[r_t \mid \mathcal{G}_{t-1}] = v_{\mathrm{bnd}}(\theta, \vartheta_t) \quad \text{whenever } \vartheta_t \le \theta. \tag{96}$$

(P5) *Informative initialization.* When $\kappa_{\underline{\theta}}$ is used in the first episode, $a_1 \neq 0$ almost surely. Hence $V_t > 0$ almost surely for every $t \ge 1$.

Thus conservative operation is itself a repeatable experiment. As the design parameter approaches the truth, its expected value approaches the oracle value.

5.2 Lower-confidence policy

For $t \ge 1$, define the weighted least-squares estimate

$$\widehat{\theta}_t := \frac{\sum_{s=1}^{t} a_s Y_s/\sigma^2}{V_t}. \tag{97}$$

For a mixture scale $\nu > 0$ and confidence level $\delta \in (0,1)$, let

$$b(v, \delta; \nu) := \left\{ \frac{2(v+\nu)\log\left(\frac{\sqrt{1+v/\nu}}{\delta}\right)}{v^2} \right\}^{1/2}, \qquad v > 0. \tag{98}$$

Lemma 5 (Information-indexed confidence sequence)**.** *Under* (93)*, for every $\delta \in (0,1)$ and $\nu > 0$,*

$$\mathbb{P}_\theta\Big(|\widehat{\theta}_t - \theta| \le b(V_t, \delta; \nu) \text{ for every } t \ge 1 \Big) \ge 1 - \delta. \tag{99}$$

The proof is in Appendix B.1. The radius is indexed by the accumulated information V_t and is of order $\sqrt{\log V_t / V_t}$ for large V_t.

The controller uses the lower endpoint of this confidence sequence:

$$\vartheta_1 := \underline{\theta}, \qquad \vartheta_t := \Big[\widehat{\theta}_{t-1} - b(V_{t-1}, \delta; \nu)\Big]_{\underline{\theta}}^{\overline{\theta}}, \quad t \ge 2, \tag{100}$$

where

$$[x]_{\underline{\theta}}^{\overline{\theta}} := \min\{\overline{\theta}, \max\{\underline{\theta}, x\}\}.$$

At episode t, it runs κ_{ϑ_t}, observes (r_t, Y_t), and updates (97).

5.3 Safe recovery theorem

The parameter-aware oracle receives $v^\star(\theta)$ per episode. In this episodic specialization, its expected regret is

$$R_T(\pi; \theta) := \sum_{t=1}^{T} \left(v^\star(\theta) - \mathbb{E}_\theta^\pi[r_t] \right). \tag{101}$$

Theorem 2 (Safe recovery from repeatable information)**.** *Suppose Assumption 1 holds. Then the policy* (100) *satisfies, for every $\theta \in [\underline{\theta}, \overline{\theta}]$,*

$$\overline{P}_{\theta,T}^{\pi}(\mathsf{Safe}_{\theta,T}^{c}) \le \delta, \tag{102}$$

and

$$R_T(\pi; \theta) \le r_{\max} + \sum_{t=2}^{T} \mathbb{E}_\theta^\pi[\omega(2b(V_{t-1}, \delta; \nu))] + r_{\max}(T-1)\delta. \tag{103}$$

Consequently, for $\eta_T \in (0,1)$, setting $\delta = \eta_T$ gives $\pi \in \Pi_{\mathrm{safe}}^{T}([\underline{\theta}, \overline{\theta}])$.

The proof is in Appendix B.2. Whenever $|\widehat{\theta}_{t-1} - \theta| \le b(V_{t-1}, \delta; \nu)$, the projection in (100) gives $\vartheta_t \le \theta$ and $\theta - \vartheta_t \le 2b(V_{t-1}, \delta; \nu)$. Thus (P2) gives safety and (P3) bounds the loss in that episode. Lemma 5 makes these inequalities simultaneous with probability at least $1 - \delta$; bounded reward controls the remaining event.

5.4 Rates from information growth

Theorem 2 converts a lower bound on V_t into a regret rate.

Corollary 5 (Polynomial information growth)**.** *Suppose Assumption 1 holds and, uniformly over $\theta \in [\underline{\theta}, \overline{\theta}]$, there are constants $c_-, c_+ > 0$ and $\alpha \in (0, 1]$ such that*

$$c_- t^\alpha \le V_t \le c_+ t \qquad \textit{almost surely for every } t \ge 1. \tag{104}$$

Assume also that $\omega(r) \le L r^q$ for some $L < \infty$ and $q > 0$. Run the policy with $\delta = \delta_T = T^{-2}$ and fixed $\nu > 0$. Then

$$\sup_{\theta \in [\underline{\theta}, \overline{\theta}]} R_T(\pi; \theta) = \begin{cases} O\big(L T^{1-\alpha q/2} (\log T)^{q/2}\big), & \alpha q < 2, \\ O\big(L (\log T)^{1+q/2}\big), & \alpha q = 2, \\ O\big(L (\log T)^{q/2}\big), & \alpha q > 2. \end{cases} \tag{105}$$

The hidden constants may depend on c_-, c_+, α, ν, q, and $r_{\max}$, but not on T or θ.

The proof is in Appendix B.3. For example, if $V_t \asymp t$ and $\omega(r) \le \ell r$, then

$$R_T(\pi; \theta) = O\Big(\ell \sqrt{T \log T}\Big). \tag{106}$$

If $V_t \asymp t$ and $\omega(r) \le L r^2$, then

$$R_T(\pi; \theta) = O\big(L (\log T)^2\big). \tag{107}$$

The information-growth rate and the value modulus jointly determine regret.

5.5 Alignment as a sufficient condition

Definition 7 (Safety–information alignment)**.** The family $\{\kappa_\vartheta\}_{\vartheta \in [\underline{\theta}, \overline{\theta}]}$ is *aligned* if the same experiment that satisfies the safety and repeatability condition (P2) also produces the observation (93). It is uniformly $(\rho, \bar{\rho})$-aligned if every execution satisfies

$$0 < \rho \le \frac{a_t^2}{\sigma^2} \le \bar{\rho} < \infty \qquad \text{almost surely.} \tag{108}$$

Under alignment, no separate excitation phase is needed: the maneuver used to preserve safety also supplies information.

Corollary 6 (Uniform alignment)**.** *Suppose Assumption 1 holds, the experiment family is uniformly $(\rho, \bar{\rho})$-aligned, and $\omega(r) \le \ell r$. Then the lower-confidence policy with $\delta = T^{-2}$ is uniformly chance-T^{-2} safe and*

$$\sup_{\theta \in [\underline{\theta}, \overline{\theta}]} R_T(\pi; \theta) = O\left(\ell \sqrt{\frac{T \log T}{\rho}}\right). \tag{109}$$

The hidden constant may depend on $\bar{\rho}$, ν, and $r_{\max}$.

5.6 Example: recursively feasible braking

Consider a vehicle approaching a wall at $p = 0$. Its fixed but unknown braking efficiency satisfies

$$\theta \in [\underline{\theta}, \overline{\theta}].$$

Each episode starts at $p(0) = -D$. The controller selects the initial speed, which is also the episode reward. Since $\theta \le \overline{\theta}$, the reward is bounded by $r_{\max} := \sqrt{2D\overline{\theta}}$. The controller then applies braking according to

$$\dot{p}(s) = v(s), \qquad \dot{v}(s) = -\theta u(s), \qquad u(s) \in [0, 1], \tag{110}$$

Safety requires $p(s) \le 0$ until the vehicle stops. A fixed reset then returns the vehicle to the same initial state for the next episode.

For known θ, maximal braking from speed v has stopping distance $v^2/(2\theta)$. The oracle entry speed is therefore

$$v^\star(\theta) = \sqrt{2D\theta}. \tag{111}$$

A conservative experiment uses

$$v(\vartheta) = \sqrt{2D\vartheta}, \tag{112}$$

applies maximal braking, and records the noisy deceleration measurement

$$y_t = \theta + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \tag{113}$$

Thus $a_t = 1$ in (93).

Lemma 6 (Braking verifies alignment)**.** *For every $\underline{\theta} \le \vartheta \le \theta \le \overline{\theta}$, the experiment* (112) *is safe under θ, returns to the common initial state after the reset, contributes Fisher information $1/\sigma^2$, and satisfies*

$$0 \le v^\star(\theta) - v(\vartheta) \le \sqrt{\frac{D}{2\underline{\theta}}}(\theta - \vartheta). \tag{114}$$

Corollary 7 (Safe oracle recovery for braking)**.** *The lower-confidence policy applied to the braking instance is uniformly chance-T^{-2} safe and satisfies*

$$\sup_{\theta \in [\underline{\theta}, \overline{\theta}]} R_T(\pi; \theta) = O\left(\sigma \sqrt{\frac{D}{\underline{\theta}}} \sqrt{T \log T} \right). \tag{115}$$

Each safe braking episode supplies one new observation, so $V_t = t/\sigma^2$. Unlike the finite-expenditure mechanism in Lemma 3, the experiment can be repeated without consuming a nonrenewable safety margin.

5.6.1 Nominal certainty equivalence at an active safety boundary

The braking example also separates statistical consistency from trajectory safety. Suppose that, after n safe braking episodes, the controller forms the projected sample mean $\widehat{\theta}_n^{\mathrm{CE}} := \left[\frac{1}{n}\sum_{s=1}^n Y_s\right]_{\underline{\theta}}^{\overline{\theta}}$ and then uses the certainty-equivalent entry speed $v_{n+1}^{\mathrm{CE}} := \sqrt{2D\widehat{\theta}_n^{\mathrm{CE}}}$. For any $\theta \in (\underline{\theta}, \overline{\theta})$, the resulting episode is safe if and only if $\widehat{\theta}_n^{\mathrm{CE}} \le \theta$, because

$$\frac{(v_{n+1}^{\mathrm{CE}})^2}{2\theta} = D\frac{\widehat{\theta}_n^{\mathrm{CE}}}{\theta}.$$

Since

$$\frac{1}{n}\sum_{s=1}^n Y_s - \theta \sim \mathcal{N}\left(0, \frac{\sigma^2}{n}\right),$$

and projection preserves comparison with an interior value of θ,

$$\mathbb{P}_\theta\left(\frac{(v_{n+1}^{\mathrm{CE}})^2}{2\theta} > D \right) = \mathbb{P}_\theta\left(\widehat{\theta}_n^{\mathrm{CE}} > \theta \right) = \frac{1}{2}$$

for every n.

More generally, consider the confidence-adjusted plug-in rule

$$\widehat{\theta}_{n,m} := \left[\frac{1}{n}\sum_{s=1}^n Y_s - m_n \right]_{\underline{\theta}}^{\overline{\theta}}, \qquad v_{n+1}(m_n) := \sqrt{2D\widehat{\theta}_{n,m}},$$

where $m_n \geq 0$. For an interior parameter,

$$\mathbb{P}_\theta\left(\frac{v_{n+1}(m_n)^2}{2\theta} > D\right) = 1 - \Phi\left(\frac{\sqrt{n}\,m_n}{\sigma}\right),$$

where Φ is the standard Gaussian distribution function. Thus a one-episode violation probability at most $\delta < 1/2$ requires

$$m_n \geq \frac{\sigma}{\sqrt{n}}\,\Phi^{-1}(1-\delta).$$

Increasing n therefore shrinks the magnitude of the certainty-equivalent overshoot, but does not shrink its probability when $m_n = 0$. A trajectory-wide guarantee requires a time-uniform, data-dependent margin, which is the role of the confidence sequence in the lower-confidence policy.

5.7 Relation to pre-commitment information

Definition 4 and Lemma 1 identify pre-commitment information with a stopped sum of one-step divergences. In the present Gaussian model, this sum is a multiple of V_T.

Proposition 2 (Pre-commitment KL for repeatable experiments)**.** *Fix $\theta \neq \theta'$ and a policy that is uniformly safe for $\{\theta, \theta'\}$ and executes experiments of the form* (93) *for T episodes without entering a model-specific commitment regime. Then*

$$D_{\mathrm{KL}}\left(P^{\pi,\mathrm{pre}}_{\theta,T} \,\middle\|\, P^{\pi,\mathrm{pre}}_{\theta',T}\right) = \frac{(\theta-\theta')^2}{2}\mathbb{E}^\pi_\theta[V_T]. \tag{116}$$

Consequently, if such a policy satisfies $V_T \geq g(T)$ almost surely, then

$$I_{\mathrm{pre}}(T;\theta,\theta') \geq \frac{(\theta-\theta')^2}{2}g(T). \tag{117}$$

Under uniform ρ-alignment, repeating the common-safe experiment $\kappa_{\underline{\theta}}$ remains noncommitting and therefore

$$I_{\mathrm{pre}}(T;\theta,\theta') \geq \frac{\rho}{2}(\theta-\theta')^2 T. \tag{118}$$

The lower bound requires bounded information for every uniformly safe policy that has not committed. Here one common-safe experiment gives linear information growth, so the bounded-information obstruction is absent. The recovery theorem also shows how to use that information through a safe confidence rule and convert estimation error into value loss.

Remark 9 (Scope). The result treats a scalar parameter with a known order in which smaller design values are conservative. It also requires repeatable experiments and a common reset or continuation state. These are sufficient conditions, not necessary ones. Vector parameters, multiple control-equivalence classes, non-resetting trajectories, and dynamic latent environments require additional transition and exploitation arguments.

6 Computing Pre-Commitment Information

Theorem 1 requires an upper bound on the information available before commitment. In general, computing this quantity is an adaptive experiment-design problem. We now give an exact reduction for a deterministic linear–Gaussian class with quadratic common-safe constraints.

This section addresses only the information calculation. The commitment event and the value lost without commitment must still be derived from the control problem. Exact formulas can also show that common-safe information grows, but growth alone does not provide the safe transition required by Section 5. For the list problem in Section 3.5, pairwise certificates give a conservative bound on the corresponding multiclass information.

6.1 A deterministic linear–Gaussian class

Fix a pair $\theta, \theta' \in \Theta_0$ and write

$$g := \theta - \theta'.$$

Let $\mathcal{U}^T_{\text{core}}$ be the set of complete action sequences that remain in the common pre-commitment regime through horizon T.

Assumption 2 (Deterministic linear–Gaussian certificate class)**.** The following hold for the pair (θ, θ').

(D1) Under $\vartheta \in \{\theta, \theta'\}$, the observations obey the sequential model

$$y_t = \vartheta^\top \zeta_t(u_{1:t}) + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \tag{119}$$

where ζ_t is a deterministic causal function, common to the two environments, and ξ_t is independent of the past and of the policy's private randomization.

(D2) Membership in $\mathcal{U}^T_{\text{core}}$ depends only on the action sequence, not on the observation noise.

(D3) Every action prefix that can occur strictly before commitment under a uniformly safe policy extends to a sequence in $\mathcal{U}^T_{\text{core}}$. Conversely, every sequence in $\mathcal{U}^T_{\text{core}}$ can be played open loop by a uniformly safe policy without commitment.

This class includes the consumable-resource examples in which the regressor is a known function of the controls. It does not cover general partially observed stochastic systems, where noise or feedback changes which future experiments remain feasible.

For $u_{1:T} \in \mathcal{U}^T_{\text{core}}$, define

$$\Phi_T(u_{1:T}) := \sum_{t=1}^{T} \big(g^\top \zeta_t(u_{1:t})\big)^2. \tag{120}$$

Proposition 3 (Open-loop information reduction)**.** *Under Assumption 2,*

$$I_{\text{pre}}(T; \theta, \theta') = \frac{1}{2\sigma^2} \sup_{u_{1:T} \in \mathcal{U}^T_{\text{core}}} \Phi_T(u_{1:T}). \tag{121}$$

Thus feedback and randomization do not increase the maximum pre-commitment information in this class. The proof is in Appendix C.1.

Equation (121) is a constrained experiment-design problem: choose a common-safe action sequence to maximize signal energy. The unconstrained counterpart is classical deterministic input design, where open-loop sequences with certified excitation can be constructed explicitly [11, 12]; the constraint that the sequence remain in $\mathcal{U}^T_{\text{core}}$ is what caps the value of the program.

6.2 Quadratic information programs

Suppose the regressors are linear in the stacked action vector

$$\mathbf{u} := \begin{bmatrix} u_1^\top & \cdots & u_T^\top \end{bmatrix}^\top \in \mathbb{R}^{n_T}, \qquad \zeta_t(\mathbf{u}) = Z_{t,T}\mathbf{u},$$

where causality means that $Z_{t,T}$ has zero columns for controls after time t. Define

$$M_T := \sum_{t=1}^{T} Z_{t,T}^\top g g^\top Z_{t,T} \succeq 0. \tag{122}$$

Then $\Phi_T(\mathbf{u}) = \mathbf{u}^\top M_T \mathbf{u}$.

Single quadratic budget. Suppose

$$\mathcal{U}_{\mathrm{core}}^{T} = \left\{\mathbf{u} : \mathbf{u}^\top K_T \mathbf{u} \le \epsilon_T\right\}, \qquad K_T \succeq 0, \quad \epsilon_T > 0. \tag{123}$$

Define the generalized Rayleigh quotient (with value zero when $M_T = 0$)

$$\lambda_{\max}(M_T, K_T) := \sup_{\mathbf{u}:\, \mathbf{u}^\top K_T \mathbf{u} > 0} \frac{\mathbf{u}^\top M_T \mathbf{u}}{\mathbf{u}^\top K_T \mathbf{u}}. \tag{124}$$

Proposition 4 (Single-budget certificate)**.** *Assume $M_T, K_T \succeq 0$.*

1. *If $\ker K_T \not\subseteq \ker M_T$, then the information program is unbounded: the model contains an informative direction with zero quadratic safety cost.*
2. *If $\ker K_T \subseteq \ker M_T$, then*

$$I_{\mathrm{pre}}(T; \theta, \theta') = \frac{\epsilon_T}{2\sigma^2} \lambda_{\max}(M_T, K_T). \tag{125}$$

The proof is in Appendix C.2.

The profile is therefore the available quadratic safety resource multiplied by the largest ratio of information to resource. Bounded information follows when

$$\epsilon_T \lambda_{\max}(M_T, K_T)$$

is uniformly bounded; linear growth follows when this product is $\Theta(T)$.

Corollary 8 (Stationary matched-order systems)**.** *Suppose*

$$M_T = I_T \otimes M_0, \qquad K_T = I_T \otimes K_0,$$

and $\ker K_0 \subseteq \ker M_0$*. Then*

$$I_{\mathrm{pre}}(T; \theta, \theta') = \frac{\epsilon_T}{2\sigma^2} \lambda_{\max}(M_0, K_0). \tag{126}$$

A bounded ϵ_T gives a bounded profile. If $\epsilon_T = \Theta(T)$ and $\lambda_{\max}(M_0, K_0) > 0$, the profile grows linearly. The proof is in Appendix C.3.

6.3 Multiple constraints and an SDP certificate

Suppose the common core is contained in the quadratic outer approximation

$$\mathcal{U}_{\mathrm{core}}^{T} \subseteq \widehat{\mathcal{U}}_{\mathrm{core}}^{T} := \left\{\mathbf{u} : \mathbf{u}^\top Q_{j,T} \mathbf{u} \le b_{j,T}, \quad j = 1, \ldots, m_T\right\}, \tag{127}$$

where $Q_{j,T} \succeq 0$ and $b_{j,T} \ge 0$. Then

$$2\sigma^2 I_{\mathrm{pre}}(T; \theta, \theta') \le v_T := \max_{\mathbf{u}} \left\{\mathbf{u}^\top M_T \mathbf{u} : \mathbf{u}^\top Q_{j,T} \mathbf{u} \le b_{j,T},\ j \in [m_T]\right\}. \tag{128}$$

The QCQP is nonconvex in general. Its semidefinite dual is

$$\begin{aligned} d_T^{\mathrm{sdp}} := \min_{\lambda \ge 0} \quad & \sum_{j=1}^{m_T} b_{j,T} \lambda_j \\ \text{subject to} \quad & \sum_{j=1}^{m_T} \lambda_j Q_{j,T} \succeq M_T. \end{aligned} \tag{129}$$

Weak duality gives $v_T \le d_T^{\mathrm{sdp}}$. Thus every feasible dual point is a valid information upper bound; strong duality is not required.

Theorem 3 (Uniform SDP information certificate)**.** *Suppose that for every horizon T there are multipliers $\lambda_{j,T} \geq 0$ such that*

$$\sum_{j=1}^{m_T} \lambda_{j,T} Q_{j,T} \succeq M_T \tag{130}$$

and

$$\sum_{j=1}^{m_T} b_{j,T} \lambda_{j,T} \leq \Lambda \qquad \text{for all } T \tag{131}$$

for some finite Λ. Then

$$\beta_{\text{pre}}(\theta, \theta') \leq \frac{\Lambda}{2\sigma^2}. \tag{132}$$

Once the commitment allowance and the noncommitment regret profile have been bounded, Theorem 1 applies with information ceiling $\Lambda/2\sigma^2$ Appendix C.4.

A finite-horizon solve bounds only $I_{\text{pre}}(T)$. A bounded-capacity certificate requires feasible multipliers whose objective values are uniformly bounded in T. Conversely, a growing SDP value does not prove that the true information profile grows: the relaxation may be loose. Even exact information growth does not by itself establish safe oracle recovery.

Remark 10 (Affine terms and input constraints). Affine dynamics, nonzero initial states, off-center ellipsoids, and linear input constraints can be represented by homogenizing with $\widetilde{\mathbf{u}} = (1, \mathbf{u}^\top)^\top$ and imposing $\widetilde{u}_0 = 1$ (or $W_{00} = 1$ after lifting). The resulting SDP remains a valid upper certificate, but the relaxation may be more conservative.

6.4 Labeled problems

Pairwise certificates also give a conservative upper bound on the mutual information used in Proposition 1. Let $\Theta_M = \{\theta_1, \ldots, \theta_M\}$ and use the same labeled stopped record S_T^π under every label. Suppose that every $\pi \in \Pi_{\text{safe}}^T(\Theta_M)$ satisfies

$$D_{\text{KL}}\Big(P_{\theta_j,T}^{\pi,\text{pre}} \,\Big\|\, P_{\theta_k,T}^{\pi,\text{pre}}\Big) \leq \overline{\beta}_{jk,T}, \qquad j,k \in [M].$$

Then

$$I_{\text{pre}}^{(M)}(T; \Theta_M) \leq \frac{1}{M^2} \sum_{j=1}^{M} \sum_{k=1}^{M} \overline{\beta}_{jk,T}. \tag{133}$$

The proof is in Appendix C.5. The bound can be loose because the pairwise programs may favor different sensing actions. For the unknown-location problem in Section 4.4, the direct list calculation is sharper.

6.5 Finite-headroom example

Recall the finite-headroom model used in the binary consequence of Theorem 1:

$$d_{t+1} = d_t - \theta u_t^2, \qquad y_t = \theta u_t + \xi_t, \qquad 0 \leq u_t \leq U.$$

For the pair (θ_b, θ_d) with $0 \leq \theta_b < \theta_d$, the common regime satisfies

$$\sum_{t=1}^{T} u_t^2 \leq \frac{d_0}{\theta_d}.$$

The information matrix is

$$M_T = (\theta_d - \theta_b)^2 I_T, \qquad K_T = I_T.$$

With the pointwise action cap, the largest feasible input energy is $\min\{TU^2, d_0/\theta_d\}$. Hence

$$I_{\mathrm{pre}}(T;\theta_b,\theta_d) = \frac{(\theta_d-\theta_b)^2}{2\sigma^2}\min\left\{TU^2, \frac{d_0}{\theta_d}\right\}. \tag{134}$$

Once $TU^2 \ge d_0/\theta_d$, the profile saturates at $(\theta_d-\theta_b)^2 d_0/(2\sigma^2\theta_d)$, which is the bound used in the finite-headroom consequence.

The one-sided budget in Remark 8 is better handled directly. If $0 \le u_t \le U$ and $\sum_t u_t \le B$, then

$$\sum_t u_t^2 \le \min\{B^2, UB\}.$$

No quadratic-budget reformulation is needed.

Scope of the computational certificate. A feasible finite-horizon dual point bounds one profile value. A uniformly bounded family of dual points certifies bounded pre-commitment information. Pairwise bounds can also be combined into a sufficient multiclass certificate. A growing SDP upper bound does not establish persistent information, and none of these calculations supplies the commitment value or the safe transition to the oracle; those remain separate control arguments.

7 Numerical Illustrations

The preceding sections give three outcomes. Bounded pre-commitment information yields an oracle-gap lower bound through Theorem 1. Repeatable information aligned with safety yields oracle recovery through Theorem 2. In the multi-alternative setting of Section 4.4, finite sensing can instead support a certified list and partial adaptation.

This section illustrates these outcomes. The replenishment and sensor examples evaluate analytic information profiles. The unknown-location example uses Monte Carlo only to evaluate the resulting list sizes and values. The braking example compares the simulated lower-confidence policy with the exact robust baseline, nominal certainty equivalence, and the analytic regret upper bound from Theorem 2. Monte Carlo is used only to display policy performance; the safety guarantee for the lower-confidence policy remains analytic. Safety in all four examples is established analytically, not by simulation. The quantities I_{av}, I_{rate}, and I_{level} are example-specific evidence quantities used in the plots; I_{pre} in Definition 4 remains the general profile.

7.1 Replenishment: from bounded to persistent safe information

Consider the replenishing thermal resource

$$d_{t+1} = \min\{d_{\max}, d_t + b - \theta u_t^2\}, \qquad d_t \ge 0, \qquad 0 \le u_t \le U, \tag{135}$$

with sensor

$$y_t = \theta u_t + \xi_t, \qquad \xi_t \overset{\mathrm{iid}}{\sim} \mathcal{N}(0,\sigma^2). \tag{136}$$

We compare the target $\theta_0 = 0$ with the incompatible alternative $\theta_1 > 0$. The θ_0-oracle uses $u_t = U$. Before excluding θ_1, the learner must preserve feasibility under θ_1.

The largest Gaussian divergence available over a length-T common-safe experiment is

$$I_{\mathrm{av}}(T;b) = \frac{\theta_1^2}{2\sigma^2}\min\left\{U^2T, \frac{d_0+bT}{\theta_1}\right\}. \tag{137}$$

The first term is the action limit; the second is the initial headroom plus replenishment. Figure 2 uses

$$\theta_0 = 0, \quad \theta_1 = 1, \quad \sigma = 1, \quad U = 1, \quad d_0 = d_{\max} = 2.$$

In the plotted horizon range, the resource term is active and

$$I_{\rm av}(T; b) = 1 + \frac{b}{2}T.$$

Thus $b = 0$ gives a bounded information budget, whereas every fixed $b > 0$ gives a positive information rate.

A robust-then-commit policy accumulates evidence inside the common-safe regime and uses $u = U$ only after the log-likelihood ratio in favor of θ_0 over θ_1 reaches

$$\kappa_T = \log(1/\eta_T) = 2\log T, \qquad \eta_T = T^{-2}. \tag{138}$$

Under θ_1, the exponentiated likelihood ratio is a nonnegative martingale. Ville's inequality therefore bounds the probability of false commitment by $e^{-\kappa_T} = T^{-2}$.

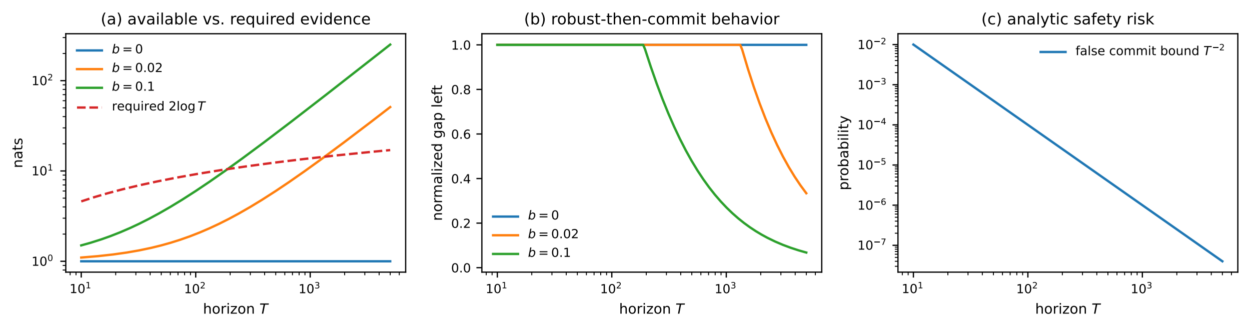


Figure 2: **Replenishment changes the information-growth regime** (Sec 7.1). Replenishing finite-headroom device with target $\theta_0 = 0$, alternative $\theta_1 = 1$, and replenishment rate b. *(a)* Available common-safe evidence $I_{\rm av}(T; b)$ against the commitment threshold $\kappa_T = 2\log T$: bounded for $b = 0$, linear for every fixed $b > 0$. *(b)* Normalized gap left, $\min\{1, \kappa_T/I_{\rm av}(T; b)\}$: the fraction of the horizon a robust-then-commit policy spends below the threshold, and hence the fraction of the commitment-only gap it forfeits. It is an evidence-budget ratio, not a simulated regret; it equals one when the threshold is unreachable and vanishes when the threshold requires a negligible fraction of the available evidence. *(c)* The Ville bound $e^{-\kappa_T} = T^{-2}$ on false commitment under θ_1; the curve is analytic, not a Monte Carlo estimate.

For $b = 0$, increasing the horizon does not increase the evidence available before commitment. For $b > 0$, the information profile grows linearly, so the bounded-information premise of Theorem 1 no longer applies.

7.2 Sensor ablation: transient evidence versus persistent imprint

We next keep the plant, objective, and safe diagnostic action sequence fixed and change only the sensor. Set $b = 0$ and retain the pair $(\theta_0, \theta_1) = (0, \theta_1)$. The learner applies a safe prefix with

$$q := \sum_{t=1}^{m} u_t^2 \le d_0/\theta_1,$$

and then sets $u_t = 0$.

With the rate sensor

$$y_t^{\rm rate} = \theta u_t + \xi_t,$$

no further information arrives after the prefix. For every $T \ge m$,

$$I_{\rm rate}(T) = \frac{\theta_1^2 q}{2\sigma^2}. \tag{139}$$

With the level sensor

$$y_t^{\text{level}} = d_t + \xi_t,$$

the same prefix leaves a persistent separation $\Delta_d = \theta_1 q$ between the two headroom levels. Independent passive measurements then give

$$I_{\text{level}}(T) = I_{\text{level}}(m) + \frac{\Delta_d^2}{2\sigma^2}(T - m), \qquad T \geq m. \tag{140}$$

Thus the rate sensor yields finite evidence, while the level sensor converts the same finite perturbation into a linearly growing observation record.

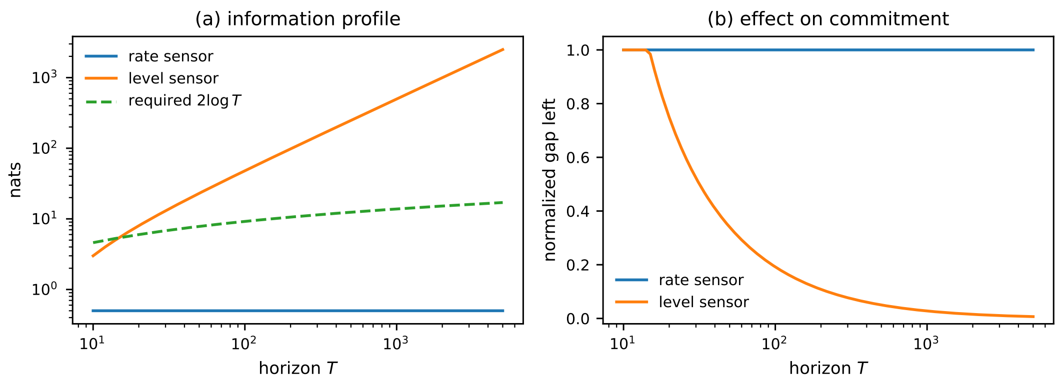


Figure 3: **The sensing channel changes the information profile** (Section 7.2); both configurations apply the same safe diagnostic prefix of energy q, and only the sensor differs. *(a)* Pre-commitment information: the rate sensor $y_t^{\text{rate}} = \theta u_t + \xi_t$ records only the transient input, giving the bounded profile; the level sensor $y_t^{\text{level}} = d_t + \xi_t$ observes the persistent headroom separation $\Delta_d = \theta_1 q$, so information grows linearly. *(b)* Normalized gap left, $\min\{1, \kappa_T / I(T)\}$, as in Figure 2(b): the rate sensor never reaches the growing threshold $\kappa_T = 2 \log T$, while the level sensor reaches it after logarithmically many passive measurements.

The example isolates the role of observation design: the plant dynamics alone do not determine the pre-commitment information regime.

7.3 Unknown disturbance location: safe lists and partial adaptation

We use the multi-alternative model from Section 4.4. There are M possible affected locations and one true location J. During the diagnostic window,

$$x_{i,t+1} = x_{i,t} + \Delta \mathbf{1}\{i = J\}, \qquad x_{i,1} = 0, \qquad x_{i,t} \leq H, \tag{141}$$

with $H = N\Delta$. Hence unit-throughput diagnostic operation is safe for N rounds. Varying N changes the available physical headroom, not only a policy parameter; the curves therefore compare different safe sensing windows. At each round, the learner inspects one location:

$$y_t = \mu \mathbf{1}\{A_t = J\} + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \tag{142}$$

At the deadline, the learner returns a nonempty certified list $\mathcal{C}_N$. The post-diagnostic controller divides mitigation and throughput across the retained locations. If $J \in \mathcal{C}_N$, it arrests the affected state. If $J \notin \mathcal{C}_N$, the state violates the constraint on the next round and the continuation reward is set to zero. Therefore safety reduces to retaining the true location.

We use the elimination rule from Section 4.4. A candidate is deleted only when its likelihood-ratio statistic exceeds $\kappa_T = \log(1/\eta_T)$. Since the lists only shrink, Ville's inequality gives, for every true location J,

$$\mathbb{P}_J(J \notin \mathcal{C}_N) \le \eta_T. \tag{143}$$

At each round, the policy inspects the least-sampled surviving location. This balanced rule is used to demonstrate partial certification; it is not claimed to be an optimal sensing policy.

All three controllers receive unit reward during the N diagnostic rounds. Afterward, the robust controller retains all M locations, the list controller uses $\mathcal{C}_N$, and the oracle retains only J. Their values are

$$J_T^{\text{robust}}(N) = N + \frac{T-N}{M}, \tag{144}$$

$$J_T^{\text{list}}(N) = N + (T-N)\mathbb{E}\left[\frac{\mathbf{1}\{J \in \mathcal{C}_N\}}{|\mathcal{C}_N|}\right], \tag{145}$$

$$J_T^{\text{oracle}}(N) = T. \tag{146}$$

The fraction of the robust-to-oracle gap recovered by the list policy is

$$\Gamma_N := \frac{J_T^{\text{list}}(N) - J_T^{\text{robust}}(N)}{J_T^{\text{oracle}}(N) - J_T^{\text{robust}}(N)}. \tag{147}$$

Panel (a) of Figure 4 reports

$$I(J; S_N) = \log M - \mathbb{E}[H(p_N)],$$

where p_N is the posterior computed from the exact Gaussian likelihoods under a uniform prior. We use

$$M = 8, \quad \mu = 2.5, \quad \sigma = 1, \quad T = 1000, \quad \eta_T = T^{-2} = 10^{-6}.$$

For each value of N, all Monte Carlo quantities are averaged over 10^5 independent draws of J and the sensor noise. The standard error of every reported reward rate or probability is therefore at most $1/(2\sqrt{10^5}) < 1.6 \times 10^{-3}$.

At $N = 36$, the reward rates are

$$\text{robust} = 0.1565, \qquad \text{safe list} = 0.3893, \qquad \text{oracle} = 1.$$

The list policy recovers 27.6% of the robust-to-oracle gap. Its mean list size is 3.62; 10.6% of trials produce a singleton list and 88.9% produce a proper partial list. No false elimination was observed in 10^5 trials. The safety claim is the analytic bound (143), not the empirical count.

This example realizes the intermediate case quantified by Proposition 1: a smaller certified uncertainty set can improve value before exact localization is possible.

7.4 Repeatable braking: safe information converted into performance

The final illustration uses the braking model of Section 5.6. The safety-preserving maneuver is also informative. We set

$$\theta \in [0.5, 1.5], \qquad \theta^\star = 1, \qquad D = 1, \qquad \sigma = 0.3.$$

Each episode begins at distance D. Given a pessimistic parameter ϑ_t, the learner enters with speed

$$v(\vartheta_t) = \sqrt{2D\vartheta_t},$$

applies maximal braking, and observes

$$y_t = \theta^\star + \xi_t.$$

Because the design coefficient is one in every episode, the accumulated information is

$$V_t = t/\sigma^2. \tag{148}$$

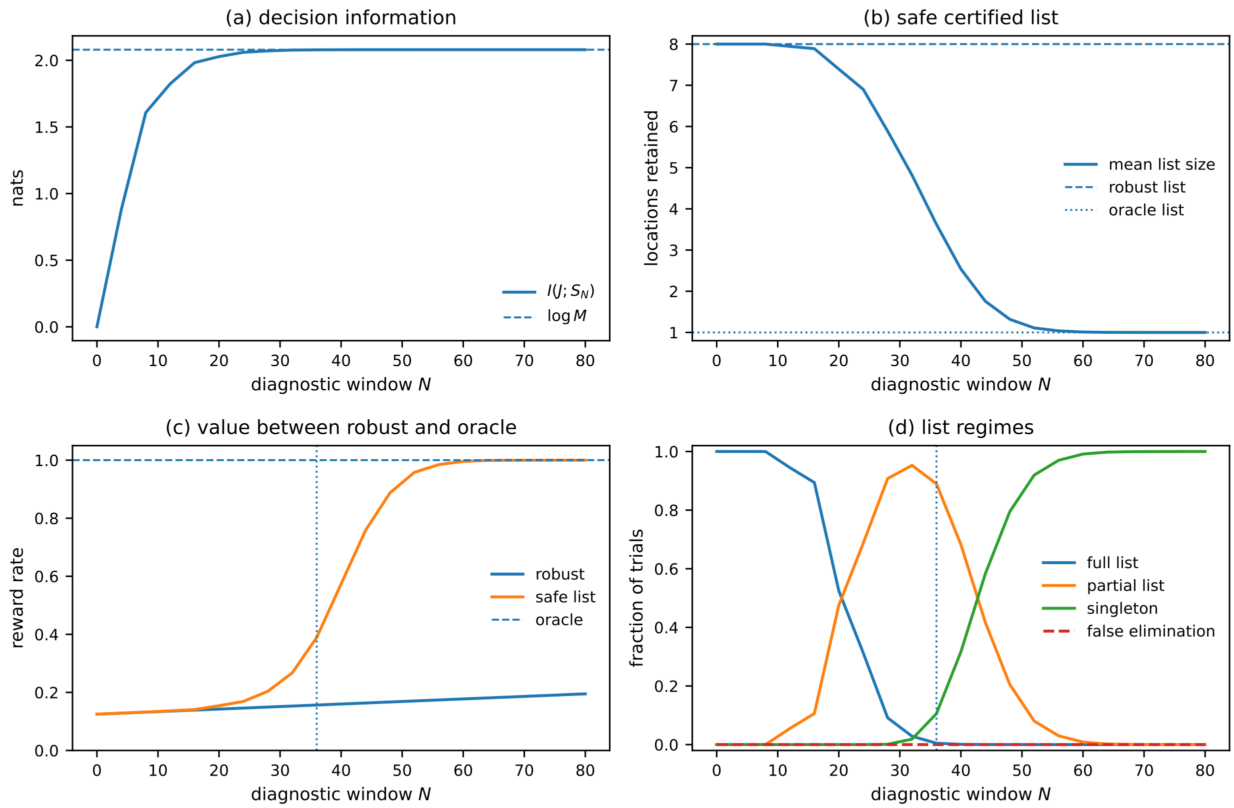


Figure 4: **Limited sensing produces safe partial adaptation.** Unknown-disturbance-location model (Section 7.3) with $M = 8$ locations, true location J, and a diagnostic window of N safe rounds; parameters as in the text. *(a)* Decision information $I(J; S_N)$ approaches $\log M$ as N grows. *(b)* Mean size of the certified list $\mathcal{C}_N$, shrinking from the robust list $[M]$ toward the oracle list $\{J\}$. *(c)* Post-diagnostic reward rates: the list policy lies between the robust and oracle values over a broad range of N. *(d)* Fraction of trials returning the full list, a proper partial list, or a singleton; near the operating point $N = 36$ most trials produce a proper partial list rather than exact localization. No false elimination occurred in 10^5 trials; the safety claim is the analytic bound, not the empirical count.

The learner uses the lower-confidence rule in Section 5.2 with $\delta_T = T^{-2}$. On the simultaneous confidence event, $\vartheta_t \leq \theta^\star$ for every t, so the stopping-distance calculation gives safety episode by episode. Corollary 7 gives

$$R_T = O\left(\sigma\sqrt{\frac{D}{\underline{\theta}}}\sqrt{T\log T}\right). \tag{149}$$

A permanently robust controller uses $v(\underline{\theta})$, so its regret at $\theta^\star > \underline{\theta}$ is linear.

We fix a terminal horizon $T_{\max} = 5000$, set $\delta = T_{\max}^{-2}$ and $\nu = 1/\sigma^2$, and simulate 20,000 independent trajectories in Figure 5. All quantities at episode $t \leq T_{\max}$ are prefixes of this single horizon-$T_{\max}$ policy. For nominal certainty equivalence, no violation through episode $t = n+1$ is equivalent to the first n centered Gaussian partial sums remaining nonpositive. For continuous symmetric increments, the corresponding survival probability is

$$\mathbb{P}_{\theta^\star}\big(\text{no CE violation through episode } t\big) = \frac{\binom{2n}{n}}{4^n}, \qquad n = t-1.$$

Hence

$$\mathbb{P}_{\theta^\star}\big(\text{CE violation by episode } t\big) = 1 - \frac{\binom{2n}{n}}{4^n}.$$

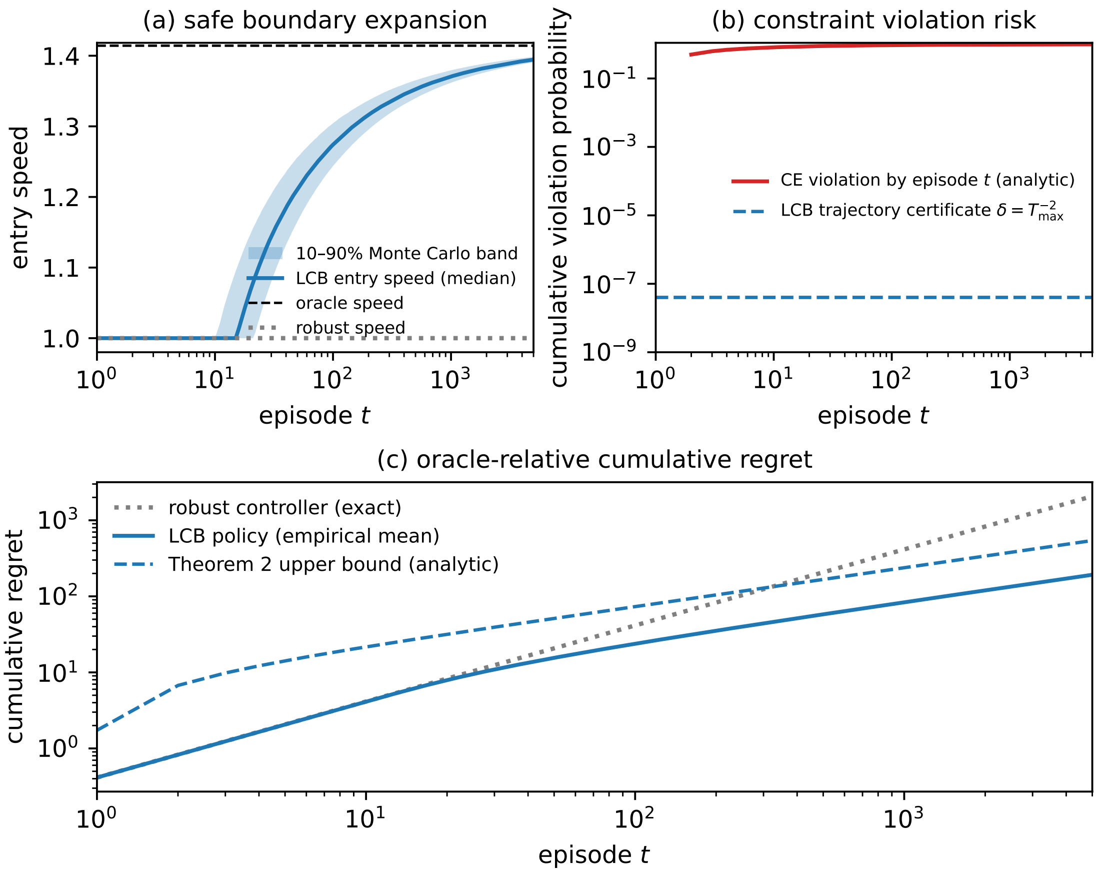


Figure 5: **Repeatable information must be converted into a safety-valid decision rule.** Recursively feasible braking with $\theta \in [0.5, 1.5]$, $\theta^\star = 1$, $D = 1$, $\sigma = 0.3$, $T_{\max} = 5000$, $\delta = T_{\max}^{-2}$, and $\nu = 1/\sigma^2$. **(a)** Median entry speed of the lower-confidence policy over 20,000 independent trajectories, with the 10–90 percentile band. The policy initially coincides with the robust controller and then expands the certified safe boundary toward the oracle. **(b)** Cumulative probability that nominal certainty equivalence has violated the stopping-distance constraint by episode t, compared with the trajectory-wide LCB certificate δ. **(c)** Exact cumulative regret of the permanently robust controller, empirical mean regret of the LCB policy, and the analytic upper bound from Theorem 2. Nominal certainty equivalence is omitted from the regret comparison because it is not uniformly safe. No LCB violations were observed in simulation; the safety claim is the analytic bound $\mathbb{P}(\text{Safe}^c) \leq \delta$.

This example verifies the two ingredients used in Section 5: one safe policy produces persistent information, and the value loss decreases with the confidence radius.

8 Related Work

Adaptive control and constrained LQR. Finite-time adaptive-control results obtain regret or sample-complexity bounds when deliberate excitation, process noise, or closed-loop state variation keeps the system informative [1, 13, 14, 15]. Certainty equivalence can be efficient in unconstrained LQR when a point estimate may be used directly for control [16]. A binding trajectory constraint changes this implication: an optimistic estimation error can itself make the selected action unsafe before later observations can correct the estimate. Section 7.4 gives a simple instance. Learning-based MPC can separate a robust model used for safety from a learned model used for performance, with convergence to the model-informed controller under sufficient excitation [17]. Constrained-LQR results give more specific guarantees. Dean et. al. [2] combine persistent excitation with robust controller synthesis. Li et. al. [3] obtain $\widetilde{O}(T^{2/3})$ regret for a single-trajectory constrained problem. Schiffer and Janson [4] show that a binding safety constraint can itself generate useful state variation in a one-dimensional problem with nonlinear baselines. Hutchinson et. al. [5] obtain $\widetilde{O}(\sqrt{T})$ regret for multidimensional chance-constrained LQR. These results give conditions under which safe operation continues to supply information. In our terms, they place the problem in a persistent precommitment-information regime. We ask whether this property holds for a given dynamics–sensor–constraint geometry. Bounded precommitment information gives a certificate that it does not.

Dual control, experiment design, and controlled sensing. Dual control accounts for the effect of an action on both current performance and future information [18, 19]. Experiment design for control selects inputs for their value to the eventual control task [20]. Sequential experimental design and controlled sensing select experiments to distinguish among hypotheses [6, 7, 8, 10]. Our setting adds a decision-time constraint. The action that uses the information must be chosen before the observation generated by that action is available. Safety may also shrink the set of admissible experiments as margin is spent. Thus full-horizon information can count evidence that arrives too late, and the relevant likelihood ratio is stopped before commitment.

Safe learning, recursive feasibility, and returnability. Safe learning and learning-based control are surveyed by García and Fernández [21], Hewing et al. [22], and Brunke et al. [23]. Recursive feasibility and controlled invariance are central tools in constrained control [24, 25]. Safety filters and reachability methods preserve a safe continuation while a learning controller operates [26, 27, 28]. Safe exploration methods use recoverability, returnability, or terminal safe sets to avoid states with no safe exit [29, 30, 31, 32]. In the deterministic systems considered here, noncommitment relative to an alternative is exactly preservation of hard-safe recursive feasibility under that alternative. Commitment is the first action that removes this continuation. Our lower bound asks whether the evidence needed for that action can be collected while the continuation is still preserved. Related methods include safe Bayesian optimization and safe linear bandits [33, 34]; safety issues in power-system learning are surveyed by Dobbe et al. [35].

Data informativity. Data informativity asks whether a given dataset supports a controller that works for every model consistent with the data. This line builds on persistency of excitation [11] and has been developed into a general informativity theory [36]. Worst-case identification bounds what a finite data record can resolve when the model class itself is uncertain [37, 38], and deterministic input designs with certified excitation show how such records can be generated [12]. Mixed stochastic–set-membership formulations sharpen the excitation story: persistency of excitation is necessary, not only sufficient, for uniform consistency, and finite-data lower bounds hold without assuming a consistent estimator [39]. Our question is online and decision-timed: whether safe interaction can generate the data needed for a model-specific decision before that decision must be made. The distinction is between certifying a controller from available data and certifying whether the required data can be collected safely in time.

9 Discussion and Limitations

When safety changes the learning problem. If a near-oracle policy can be reached while preserving a safe continuation under every plausible model, the safety-specific obstruction is absent. Information can continue to grow inside the common feasible regime, and the problem is in the setting addressed by constrained adaptive-control analyses. The lower-bound examples have a different geometry. Better target performance requires leaving the common regime, while the actions that separate the target from an alternative consume the margin that must be preserved under that alternative. The issue is then not only the rate at which uncertainty shrinks. It is whether the required evidence is available before the first action that needs it.

Information timing versus information growth. Persistent excitation asks whether information continues to grow along the closed loop. Precommitment information asks whether the relevant information arrives in time. A system may be identifiable from its complete trajectory and still fail to support safe adaptation if the observations that distinguish the relevant models appear only after commitment. Stopping the record is therefore part of the causal statement, not only a proof technique. The positive theorem treats a case in which the safe maneuver itself is informative and repeatable, so information accumulates without giving up recursive feasibility.

Design implications. When persistent safe information is available, it must still be converted into a safety-valid decision rule. In the braking example, nominal certainty equivalence can overestimate braking authority and violate the constraint, whereas the lower-confidence rule expands the safe operating boundary only as the data justify it. When precommitment information is bounded, however, changing the estimator cannot remove the obstruction; the controlled experiment or the required degree of specialization must change. Replenishment, a persistent sensor signal, and certified candidate lists illustrate these three possibilities.

Limits of the present results. Bounded precommitment information is a sufficient certificate for an unavoidable oracle gap. Failure of the certificate does not prove recovery. The pairwise profile may be large because different policies are informative for different alternatives, even though no single safe policy resolves all of them. Information may also be available but unusable because the controller cannot enter the selected regime without losing feasibility. The recovery theorem proves one positive result for an ordered scalar parameter, a repeatable experiment family, and a common continuation state. Extending it to vector parameters, several control-equivalence classes, and nonresetting dynamics requires additional transition and exploitation arguments. The computational certificate omits process noise, and the general case of unbounded but sublinear precommitment information remains unresolved.

The lower-bound theorem itself is more general. An application must specify a predictable commitment rule, compute or bound the allowed commitment probability under the alternative, evaluate the noncommitment regret profile under the target, and bound the stopped information. Pathwise value ceilings are one way to control the regret profile in deterministic systems; they are not required by the theorem.

Acknowledgments

The author thanks Aditya Gangrade and Alex Olshevsky for technical discussions, and Emiliano Dall'Anese for help with practical applications. Large language models were used for editorial assistance, including identifying possible issues in exposition, consistency, and references. The author independently developed and verified all mathematical arguments and conclusions, checked the cited sources, and assumes full responsibility for the manuscript.

A Proofs for Section 4

A.1 Proof of Lemma 3

Proof. Fix a uniformly safe policy and one sample path. Define

$$s_t := \mathbf{1}\{t < \tau^\pi\} S_t.$$

Then $s_t \geq 0$ and, by (52),

$$\sum_{t=1}^{T} s_t \leq B_T.$$

For $q \geq 1$,

$$\sum_{t=1}^{T} s_t^q \leq \left(\sum_{t=1}^{T} s_t \right)^q \leq B_T^q.$$

Using (53),

$$\sum_{t=1}^{T} \mathbf{1}\{t < \tau^\pi\} \mathrm{kl}_t^{\theta,\theta'} \le \kappa \sum_{t=1}^{T} s_t^q \le \kappa B_T^q.$$

Take expectation under θ, apply Lemma 1, and then take the supremum over uniformly safe policies. This gives (54). $\square$

A.2 Proof of Lemma 4

Proof. Fix a uniformly safe policy π. The policy action at time t, the decision to continue, and the eventual specialization label are functions of the previously observed record and environment-independent private randomization. They therefore add no information about J conditional on that record. The chain rule for the stopped record gives

$$I_\pi(J; S_T^\pi) = \sum_{t=1}^{T} P_\pi(t < \tau^\pi) I_\pi(J; Z_t \mid H_{t-1}, U_t, t < \tau^\pi).$$

Only label-dependent observations contribute. By assumption there are at most N_T such observations before specialization or fallback, and each contributes at most c_{obs}. Hence

$$I_\pi(J; S_T^\pi) \le N_T c_{\mathrm{obs}}.$$

Taking the supremum over uniformly safe policies proves (76).

For the Gaussian sensor, condition on (H_{t-1}, A_t) and define

$$B_t := \mathbf{1}\{A_t = J\}.$$

The observation depends on J only through B_t, so

$$I(J; Z_t \mid H_{t-1}, A_t) = I(B_t; Z_t \mid H_{t-1}, A_t) \le H(B_t \mid H_{t-1}, A_t) \le \log 2.$$

Let

$$p_t := P(J = A_t \mid H_{t-1}, A_t), \qquad P_1 := \mathcal{N}(\mu, \sigma^2), \qquad P_0 := \mathcal{N}(0, \sigma^2).$$

The output mixture minimizes the weighted average relative entropy over the choice of reference distribution. Taking P_0 as a reference gives

$$I(B_t; Z_t \mid H_{t-1}, A_t) \le p_t D_{\mathrm{KL}}(P_1 \| P_0) \le \frac{\mu^2}{2\sigma^2}.$$

Combining the two bounds proves (78). $\square$

A.3 Proof of Proposition 1

Proof. Let

$$E := \{J \notin C(S) \text{ or } |C(S)| > L\}, \qquad P(E) = \varepsilon_L.$$

On E^c, the true label belongs to a list of size at most L, so

$$H(J \mid S, E^c) \le \log L.$$

On E, use the bound $H(J \mid S, E) \le \log M$. Therefore

$$\begin{aligned} H(J \mid S) &\le H(\mathbf{1}_E \mid S) + H(J \mid S, \mathbf{1}_E) \\ &\le h_2(\varepsilon_L) + (1 - \varepsilon_L) \log L + \varepsilon_L \log M. \end{aligned}$$

Since J is uniform on $[M]$,

$$I(J;S) = H(J) - H(J \mid S) \geq (1-\varepsilon_L)\log\frac{M}{L} - h_2(\varepsilon_L),$$

which proves (49).

If $P(J \notin C(S)) \leq \eta$ and $P(|C(S)| > L) \leq \delta$, then $\varepsilon_L \leq \eta + \delta$ by the union bound. For $L < M$, the function

$$x \longmapsto (1-x)\log(M/L) - h_2(x)$$

is decreasing on $[0, 1/2]$. Substituting $\varepsilon_L \leq \eta + \delta \leq 1/2$ gives (50). □

A.4 Safe-list guarantee and value calculation for Section 4.4

Proof of the safe-list guarantee. Fix the true location J. At a time s with $A_s = J$, the multiplicative increment in $\exp(\Lambda_{J,t})$ is

$$\frac{\phi_{0,\sigma}(Z_s)}{\phi_{\mu,\sigma}(Z_s)}.$$

Under the true model, $Z_s \sim \mathcal{N}(\mu, \sigma^2)$ conditionally on the past, so this increment has conditional expectation one. At times with $A_s \neq J$, the process does not change. Hence $\{\exp(\Lambda_{J,t})\}_{t\geq 0}$ is a nonnegative martingale under the true location, even though the sampled locations are chosen adaptively. Ville's inequality gives

$$P_J\left(\sup_{t\leq N} \Lambda_{J,t} \geq \log\frac{1}{\eta_T}\right) \leq \eta_T.$$

The event on the left is exactly false elimination of the true location before the deadline. If an empty candidate set is replaced by $[M]$, the failure probability can only decrease. This proves (88). □

Proof of the value formulas. Suppose first that $J \in \mathcal{C}_N$. For the affected component,

$$x_{J,t+1} - x_{J,t} = \frac{\Delta}{|\mathcal{C}_N|} - \frac{\Delta}{|\mathcal{C}_N|} = 0.$$

Each retained unaffected component decreases by $\Delta/|\mathcal{C}_N|$, and each nonretained unaffected component is unchanged. Thus the post-diagnostic controller is safe whenever the list contains the true location. If $J \notin \mathcal{C}_N$, the affected component starts the post-diagnostic phase at the boundary and increases on the next round, so a violation occurs immediately.

Every policy earns unit reward in each of the N diagnostic rounds. The robust controller retains all M locations and earns post-diagnostic reward $1/M$ per round. The list controller earns $1/|\mathcal{C}_N|$ per post-diagnostic round when $J \in \mathcal{C}_N$ and zero after failure. The oracle retains the singleton $\{J\}$ and earns unit reward throughout. These calculations give (89)–(91). □

B Proofs for Section 5

B.1 Proof of Lemma 5

Proof. Define

$$M_t := \sum_{s=1}^{t} \frac{a_s \xi_s}{\sigma^2}.$$

Then

$$\widehat{\theta}_t - \theta = \frac{M_t}{V_t}.$$

Because a_t is predictable and $\xi_t \sim \mathcal{N}(0, \sigma^2)$ is independent of the past, for every fixed $\lambda \in \mathbb{R}$,

$$L_t(\lambda) := \exp\left(\lambda M_t - \frac{\lambda^2}{2} V_t\right)$$

is a nonnegative martingale with initial value one.

Integrate $L_t(\lambda)$ against the Gaussian density $\lambda \sim \mathcal{N}(0, \nu^{-1})$. Completing the square gives the nonnegative martingale

$$\mathcal{M}_t = \sqrt{\frac{\nu}{V_t + \nu}} \exp\left(\frac{M_t^2}{2(V_t + \nu)}\right).$$

Ville's inequality implies

$$\mathbb{P}_\theta\left(\sup_{t \geq 1} \mathcal{M}_t \geq \frac{1}{\delta}\right) \leq \delta.$$

On the complementary event, simultaneously for all $t \geq 1$,

$$M_t^2 \leq 2(V_t + \nu) \log\left(\frac{\sqrt{1 + V_t/\nu}}{\delta}\right).$$

Dividing by V_t^2 proves

$$|\widehat{\theta}_t - \theta| \leq b(V_t, \delta; \nu)$$

for every $t \geq 1$. □

B.2 Proof of Theorem 2

Proof. Let

$$\mathcal{E}_\infty := \left\{ |\widehat{\theta}_s - \theta| \leq b(V_s, \delta; \nu) \text{ for every } s \geq 1 \right\}.$$

Lemma 5 gives $\mathbb{P}_\theta(\mathcal{E}_\infty^c) \leq \delta$.

On $\mathcal{E}_\infty$, the selected design is conservative. At $t = 1$, $\vartheta_1 = \underline{\theta} \leq \theta$. For $t \geq 2$,

$$\widehat{\theta}_{t-1} - b(V_{t-1}, \delta; \nu) \leq \theta.$$

Clipping to $[\underline{\theta}, \overline{\theta}]$ therefore preserves $\vartheta_t \leq \theta$. The same confidence bound also gives

$$0 \leq \theta - \vartheta_t \leq 2b(V_{t-1}, \delta; \nu), \qquad t \geq 2.$$

If the lower clipping is inactive, this follows directly from the two-sided confidence interval. If it is active, then $\widehat{\theta}_{t-1} - b(V_{t-1}, \delta; \nu) < \underline{\theta}$ and $\widehat{\theta}_{t-1} \geq \theta - b(V_{t-1}, \delta; \nu)$, which imply the same bound.

By Assumption 1(P2), every episode is safe and returns to the common continuation state on $\mathcal{E}_\infty$. Hence

$$\overline{P}^\pi_{\theta,T}(\mathsf{Safe}^c_{\theta,T}) \leq \mathbb{P}_\theta(\mathcal{E}_\infty^c) \leq \delta.$$

For regret, the first episode contributes at most $r_{\max}$. For $t \geq 2$, define the predictable event

$$\mathcal{E}_{t-1} := \left\{ |\widehat{\theta}_{t-1} - \theta| \leq b(V_{t-1}, \delta; \nu) \right\}.$$

It is $\mathcal{G}_{t-1}$-measurable, and $\mathbb{P}_\theta(\mathcal{E}_{t-1}^c) \leq \delta$. On $\mathcal{E}_{t-1}$,

$$\vartheta_t \leq \theta, \qquad 0 \leq \theta - \vartheta_t \leq 2b(V_{t-1}, \delta; \nu).$$

Assumption 1(P3)–(P4) then gives

$$\left(v^\star(\theta) - \mathbb{E}_\theta[r_t \mid \mathcal{G}_{t-1}]\right) \mathbf{1}_{\mathcal{E}_{t-1}} \leq \omega(2b(V_{t-1}, \delta; \nu))\, \mathbf{1}_{\mathcal{E}_{t-1}}.$$

On $\mathcal{E}_{t-1}^c$, the one-episode loss is at most $r_{\max}$. Taking expectations yields

$$\mathbb{E}^\pi_\theta[v^\star(\theta) - r_t] \leq \mathbb{E}^\pi_\theta[\omega(2b(V_{t-1}, \delta; \nu))] + r_{\max}\delta.$$

Summing over $t = 2, \ldots, T$ and adding the first-episode bound proves (103). □

B.3 Proof of Corollary 5

Proof. Set $\delta_T = T^{-2}$. Under (104), for every $1 \le t \le T$,

$$b(V_t, \delta_T; \nu) \le C_0 t^{-\alpha/2} \sqrt{\log T},$$

where C_0 depends only on c_-, c_+, α, and ν. Since $\omega(r) \le Lr^q$,

$$\omega(2b(V_t, \delta_T; \nu)) \le C_1 L (\log T)^{q/2} t^{-\alpha q/2}.$$

Theorem 2 therefore gives

$$R_T(\pi; \theta) \le r_{\max} + C_1 L (\log T)^{q/2} \sum_{t=1}^{T-1} t^{-\alpha q/2} + \frac{r_{\max}}{T}.$$

The three cases in (105) follow from the standard partial-sum bounds for $t^{-\alpha q/2}$. □

B.4 Proof of Corollary 6

Proof. Uniform alignment gives

$$\rho t \le V_t \le \bar{\rho} t.$$

Apply Corollary 5 with $\alpha = 1$, $c_- = \rho$, $c_+ = \bar{\rho}$, and $q = 1$. □

B.5 Proofs for the braking example

Proof of Lemma 6. Under maximal braking, the stopping distance from $v(\vartheta) = \sqrt{2D\vartheta}$ is

$$\frac{v(\vartheta)^2}{2\theta} = D\frac{\vartheta}{\theta} \le D.$$

Thus the vehicle stops before the wall whenever $\vartheta \le \theta$. The observation $Y_t = \theta + \xi_t$ has coefficient $a_t = 1$ and contributes Fisher information $1/\sigma^2$.

Also,

$$\begin{aligned} v^\star(\theta) - v(\vartheta) &= \sqrt{2D}\,\frac{\theta - \vartheta}{\sqrt{\theta} + \sqrt{\vartheta}} \\ &\le \sqrt{\frac{D}{2\underline{\theta}}}(\theta - \vartheta). \end{aligned}$$

This proves (114). □

Proof of Corollary 7. Lemma 6 verifies Assumption 1 with

$$\rho = \bar{\rho} = \frac{1}{\sigma^2}, \qquad \omega(r) = \sqrt{\frac{D}{2\underline{\theta}}}\, r, \qquad r_{\max} = \sqrt{2D\bar{\theta}}.$$

Apply Corollary 6. □

B.6 Proof of Proposition 2

Proof. Conditioned on $(\mathcal{G}_{t-1}, a_t)$, the two observation laws are

$$\mathcal{N}(\theta a_t, \sigma^2) \qquad \text{and} \qquad \mathcal{N}(\theta' a_t, \sigma^2).$$

Their conditional KL divergence is

$$\frac{(\theta - \theta')^2 a_t^2}{2\sigma^2}.$$

The policy does not commit during the T experiments, so Lemma 1 gives

$$D_{\mathrm{KL}}\Big(P_{\theta,T}^{\pi,\mathrm{pre}} \,\Big\|\, P_{\theta',T}^{\pi,\mathrm{pre}}\Big) = \frac{(\theta-\theta')^2}{2}\mathbb{E}_\theta^\pi\left[\sum_{t=1}^{T}\frac{a_t^2}{\sigma^2}\right],$$

which is (116). Any one uniformly safe policy gives a lower bound on the profile, proving (117). Under uniform ρ-alignment, repeating $\kappa_{\underline{\theta}}$ is safe for the full interval and gives $V_T \geq \rho T$, which proves (118). $\square$

C Proofs for Section 6

C.1 Proof of Proposition 3

Proof. Fix a uniformly safe causal policy. By Lemma 1 and Assumption 2(D1),

$$D_{\mathrm{KL}}\Big(P_{\theta,T}^{\pi,\mathrm{pre}} \,\Big\|\, P_{\theta',T}^{\pi,\mathrm{pre}}\Big) = \frac{1}{2\sigma^2}\mathbb{E}_\theta^\pi\left[\sum_{t<\tau^\pi}\big(g^\top \zeta_t(U_{1:t})\big)^2\right].$$

Every realized pre-commitment action prefix extends, by (D3), to a sequence in $\mathcal{U}_{\mathrm{core}}^T$. Since all summands are nonnegative,

$$\sum_{t<\tau^\pi}\big(g^\top \zeta_t(U_{1:t})\big)^2 \leq \sup_{u_{1:T}\in\mathcal{U}_{\mathrm{core}}^T} \Phi_T(u_{1:T})$$

on every path. Taking the supremum over uniformly safe policies gives the upper bound in (121).

Conversely, (D3) states that every sequence in $\mathcal{U}_{\mathrm{core}}^T$ can be played open loop by a uniformly safe, noncommitting policy. For this policy, the stopped record is the full horizon record, and the preceding identity gives KL divergence $\Phi_T(u_{1:T})/(2\sigma^2)$. Taking the supremum over core sequences proves the reverse inequality. $\square$

C.2 Proof of Proposition 4

Proof. Suppose first that $\ker K_T \not\subseteq \ker M_T$. Then there is a vector $z \in \ker K_T$ with $z^\top M_T z > 0$. For every $c > 0$, the vector cz is feasible, while

$$(cz)^\top M_T (cz) = c^2 z^\top M_T z \longrightarrow \infty.$$

Hence the information program is unbounded.

Now suppose $\ker K_T \subseteq \ker M_T$. Decompose $\mathbf{u} = u_r + u_0$, where $u_r \in \mathrm{range}(K_T)$ and $u_0 \in \ker K_T$. Since $M_T u_0 = 0$, both the objective and the constraint depend only on u_r. Set

$$w := K_T^{1/2} u_r.$$

Then

$$u_r = K_T^{\dagger/2} w, \qquad u_r^\top K_T u_r = \|w\|_2^2,$$

and

$$u_r^\top M_T u_r = w^\top K_T^{\dagger/2} M_T K_T^{\dagger/2} w.$$

Maximizing over $\|w\|_2^2 \le \epsilon_T$ gives

$$\sup_{\mathbf{u}^\top K_T \mathbf{u} \le \epsilon_T} \mathbf{u}^\top M_T \mathbf{u} = \epsilon_T \lambda_{\max}(M_T, K_T).$$

Divide by $2\sigma^2$ and apply Proposition 3. □

C.3 Proof of Corollary 8

Proof. On the range of K_T,

$$K_T^{\dagger/2} M_T K_T^{\dagger/2} = I_T \otimes \left(K_0^{\dagger/2} M_0 K_0^{\dagger/2}\right).$$

Its largest eigenvalue is therefore $\lambda_{\max}(M_0, K_0)$. Substituting this equality into (125) gives (126). □

C.4 Proof of Theorem 3

Proof. Fix T and a vector $\mathbf{u}$ in the quadratic outer core. By (130),

$$\begin{aligned}\mathbf{u}^\top M_T \mathbf{u} &\le \sum_{j=1}^{m_T} \lambda_{j,T} \mathbf{u}^\top Q_{j,T} \mathbf{u} \\ &\le \sum_{j=1}^{m_T} \lambda_{j,T} b_{j,T} \le \Lambda.\end{aligned}$$

Hence the QCQP value in (128) is at most Λ. Proposition 3 then gives

$$I_{\mathrm{pre}}(T; \theta, \theta') \le \frac{\Lambda}{2\sigma^2}$$

for every T. Taking the supremum over horizons proves (132). □

C.5 Proof of the labeled aggregation bound

Proof. Fix a uniformly safe policy and write

$$P_j := P^{\pi,\mathrm{pre}}_{\theta_j,T}, \qquad \overline{P} := \frac{1}{M}\sum_{k=1}^{M} P_k.$$

Under the uniform prior,

$$I(J; S_T^\pi) = \frac{1}{M}\sum_{j=1}^{M} D_{\mathrm{KL}}(P_j \| \overline{P}).$$

Convexity of KL divergence in its second argument gives

$$D_{\mathrm{KL}}(P_j \| \overline{P}) \le \frac{1}{M}\sum_{k=1}^{M} D_{\mathrm{KL}}(P_j \| P_k).$$

Applying the assumed pairwise bounds and averaging over j yields

$$I(J; S_T^\pi) \le \frac{1}{M^2}\sum_{j=1}^{M}\sum_{k=1}^{M} \overline{\beta}_{jk,T}.$$

Finally, take the supremum over uniformly safe policies to obtain (133). □

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